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Mapping the evolving landscape of AI literacy research: an integrative review combining bibliometric analysis and thematic synthesis

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ABSTRACT

Although AI literacy research has proliferated in recent years, the field remains fragmented across disciplines, methodologies, and educational contexts, limiting coherent knowledge accumulation. This study addresses this fragmentation through an integrative review combining bibliometric analysis with thematic synthesis, examining 838 peer-reviewed studies from Scopus and Web of Science. Bibliometric analysis using CiteSpace and VOSviewer reveals three developmental phases: formative, acceleration, and exponential expansion, with the 2024-2026 period alone accounting for over 90% of the corpus, reflecting the post-ChatGPT diffusion of GenAI technologies. Research production is geographically concentrated, with the United States, mainland China, Hong Kong, and Germany collectively contributing 51.4% of global output, while Africa, Latin America, and South Asia together remain markedly underrepresented. Thematic synthesis identifies 'educational strategies and curriculum design' (31.6%), 'relational studies of AI literacy with adjacent constructs' (30.5%), and 'theoretical framework development' (15.5%) as top three research foci. The field is predominantly empirical (80.9%), with quantitative (55.6%), mixed-methods (26.4%), and qualitative (18.0%) approaches represented among empirical studies. Critical gaps including framework proliferation without theoretical consolidation, the unresolved theoretical relationship between AI literacy and GenAI literacy, near-universal reliance on cross-sectional self-report designs, and underrepresentation of early childhood, vocational, and Global South contexts, require significant future research efforts.

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AI literacy; integrative review; bibliometric analysis; thematic synthesis; generative AI; large language models

1. Introduction

1.1. The imperative of AI literacy in the AI era

The proliferation of AI technologies has precipitated unprecedented transformations across economic, social, and educational domains, fundamentally reshaping how individuals interact with technological systems in their daily lives (Crawford 2021; Zhai et al. 2021). From algorithmic decision-making in healthcare and finance to AI-powered content curation on social media platforms, AI has become deeply embedded in the fabric of contemporary society (Dwivedi et al. 2021). This technological revolution extends beyond mere automation; it represents a paradigm shift that demands new forms of literacy for effective civic participation, professional competence, and personal agency in an increasingly AI-mediated world (Jarrahi 2018; Long and Magerko 2020; Ng et al. 2021).

The urgency of developing robust AI literacy stems from multiple interconnected imperatives. At the societal

level, AI systems increasingly influence critical decisions affecting individuals' lives, including credit assessments, employment screening, medical diagnoses, and criminal justice outcomes (Lepri et al. 2018; Mittelstadt et al. 2016). Without adequate AI literacy, citizens lack the capacity to critically evaluate these algorithmic determinations and advocate for transparency and accountability (Cheong 2024; Pramod 2025). Furthermore, the rise of GenAI technologies and large language models (LLMs) has introduced unprecedented challenges regarding information authenticity, content provenance, and the potential for AI-enabled misinformation, making critical AI literacy essential for navigating the contemporary information ecosystem (Chu-Ke and Dong 2024).

From an economic and workforce perspective, AI literacy has emerged as a critical determinant of employability and professional advancement. Research indicates that approximately 27–30% of current working hours in Europe and the United States could be

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automated by 2030, with millions of workers anticipated to require occupational transitions across sectors (Hazan et al. 2024). This transformation creates both threats and opportunities: while AI may eliminate certain job categories, it simultaneously generates demand for workers capable of collaborating with AI systems, interpreting AI outputs, and applying AI tools to enhance productivity (Brynjolfsson, Li, and Raymond 2025). Consequently, AI literacy represents not only a specialised technical skill but a foundational competency increasingly prerequisite for economic participation across diverse professional domains (Chee, Ahn, and Lee 2025).

Within educational contexts, the imperative for AI literacy education has gained particular prominence as AI technologies increasingly permeate learning environments through intelligent tutoring systems, automated assessment tools, and AI-powered educational platforms (Crompton and Burke 2023). Today's young learners are growing up in an environment where AI-powered applications increasingly shape their educational experiences, recreational activities, and everyday services, marking a significant generational shift in human-technology interaction (UNICEF 2021). Preparing young generation to critically engage with AI technologies, understanding both their capabilities and limitations, has become an essential educational objective that spans all levels from early childhood to adult and professional education (Laupichler et al. 2022; Sperling et al. 2024; Su, Ng, and Chu 2023).

Against this backdrop, AI literacy has emerged as a critical competency framework for navigating increasingly AI-mediated environments (Long and Magerko 2020; Ng et al. 2021). While definitions vary across disciplines, AI literacy generally encompasses the knowledge, skills, and attitudes required to critically understand, effectively utilise, and ethically engage with AI systems (Ng et al. 2021; B. Wang, Rau, and Yuan 2023). This multidimensional construct extends beyond mere technical proficiency to include critical evaluation of AI outputs, understanding of AI's societal implications, and awareness of ethical considerations surrounding AI deployment (Kong, Cheung, and Zhang 2023). The advent of GenAI technologies has further expanded the scope of AI literacy, introducing novel challenges regarding human-AI collaboration, output evaluation, and responsible implementation (Chen, Tallant, and Selig 2025; Simms 2025).

1.2. Research aims and questions

Despite the critical importance of AI literacy, systematic understanding of the research landscape remains

fragmented. The field has witnessed exponential growth, particularly following the widespread adoption of GenAI, yet this rapid expansion has occurred across disparate disciplinary traditions, educational contexts, and geographical regions. Existing research syntheses, while valuable, exhibit significant limitations in scope and methodological approach. Bibliometric analyses have mapped publication patterns but lack substantive theoretical analysis (Berigel and Şilbir 2024; Tenório et al. 2023). Traditional reviews have provided rigorous synthesis within specific educational levels or thematic foci but fail to capture the evolutionary trajectories of the field (Laupichler et al. 2022; Su, Ng, and Chu 2023). Moreover, many existing reviews predate the GenAI revolution, limiting their contemporary relevance.

To address these gaps, this study adopts an integrative approach combining bibliometric analysis and thematic synthesis to examine the knowledge structure and developmental trajectories of AI literacy research. This dual methodology of synthesis leverages the complementary strengths of bibliometric mapping and thematic synthesis. Bibliometric analysis excels at providing publication trends, collaboration networks, and keyword evolution; whereas thematic synthesis enables nuanced analysis of study findings, methodological approaches, and research context (Chen 2006; Hallinger and Kovačević 2021). The integrated approach is particularly suited for AI literacy research, an emerging, rapidly evolving, and inherently interdisciplinary domain spanning education, computer science, psychology, and policy studies that exhibit both complexity and conceptual fragmentation requiring multiple analytical lenses.

The overarching research question (RQ) which guided this integrative review was: How has AI literacy evolved as a research field, and what are the existing knowledge structures and thematic patterns that characterise this research field?

This overarching RQ is operationalised through two sub-RQs, with RQ1 being answered by bibliometric analysis and RQ2 by systematic synthesis:

RQ1: What are the developmental trajectories and structural characteristics of AI literacy research in terms of six elements: temporal trends, geographical distribution, institutional collaboration, publication venues, co-authorship networks, and keyword evolution?

RQ2: What thematic patterns emerge from AI literacy research regarding research foci, methodological approaches, target populations, and educational contexts?

By addressing these questions through bibliometric analysis and thematic synthesis, this integrative review

makes several contributions to the field. First, it provides a broad mapping of AI literacy's intellectual genealogy and evolutionary trajectories, illuminating how knowledge has accumulated across disciplinary boundaries. Second, it identifies patterns in research focus, methodological approaches, and theoretical frameworks that reveal both established conventions and emerging innovations in AI literacy research. Third, it synthesises core thematic content across educational contexts and population segments, highlighting common elements while acknowledging contextual variations. Fourth, it identifies critical research gaps and promising directions for future investigation, providing an agenda for researchers seeking to advance AI literacy scholarship across diverse contexts.

It should be noted that this review specifically focuses on AI literacy research since 'AI literacy' emerged and consolidated as a distinct research construct with dedicated conceptual frameworks (Long and Magerko 2020). While AI-related education research existed prior to this period, the formalisation of AI literacy as a coherent concept with systematic frameworks distinguishes this contemporary body of work from earlier, more diffuse investigations of AI education (Long and Magerko 2020; Ng et al. 2021). This temporal scope enables focused examination of the field's development during its most dynamic period of growth while acknowledging the broader educational lineage from which it emerges.

2. Literature review

2.1. The evolving concept of literacy in the digital age

The concept of literacy has undergone substantial transformation since its original association with written text and the ability to comprehend, employ, and express oneself through written language forms (Cazden et al. 1996; Kress 2003). The expansion of digital technologies has fundamentally shifted assumptions about literacy, rendering it a flexible and evolving concept aligned with both socio-technical practices across time, place, and space (Jewitt 2008; Leu et al. 2017). This conceptual evolution has generated a constellation of domain-specific literacies, each addressing competencies associated with a particular technological construct.

Digital literacy, emerging in the 1990s, established foundational frameworks for understanding technology-mediated competencies (Bawden 2001; Eshet 2004). Subsequently, computational literacy extended these frameworks to encompass computational reasoning and problem-solving skills (DiSessa 2018; Wing

2006). Data literacy addressed the increasing importance of understanding, interpreting, and communicating with data (Calzada Prado and Marzal 2013; Marchant and Eliason 2024). More recently, algorithmic literacy has emerged to address citizens' understanding of algorithmic systems that increasingly mediate social interactions and decision-making processes (Dogruel, Masur, and Joeckel 2022; Ridley and Pawlick-Potts 2021).

AI literacy represents the latest iteration in this evolutionary trajectory, responding to the pervasive integration of AI systems in everyday life. Understanding its relationship to antecedent literacies is essential for establishing clear conceptual boundaries. While AI literacy shares foundational elements with digital, computational, and algorithmic literacies, it extends beyond these constructs by addressing AI-specific competencies including recognising AI applications, evaluating AI outputs, and navigating the ethical implications of AI deployment (Long and Magerko 2020; Ng et al. 2021). This distinction is particularly salient given that AI systems increasingly operate with degrees of autonomy that distinguish them from earlier digital technologies.

2.2. Conceptualising AI literacy: definitions and frameworks

The conceptualisation of AI literacy has evolved considerably since its emergence as a distinct construct. Long and Magerko's (2020) seminal definition characterises AI literacy as 'a set of competencies that enables individuals to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a tool online, at home, and in the workplace' (2). This definition has achieved substantial influence in subsequent research, establishing a competency-based approach that emphasises functional, critical, and applied dimensions of AI understanding.

Alternative conceptualizations have emerged across different domains and target populations. Kong and Zhang (2021) define AI literacy as 'understanding of AI concepts and competencies in using AI concepts for evaluation and using AI concepts for understanding the real world' (12), emphasising cognitive and applied dimensions. X. Wang, Li, and Huang (2023) adopt an information literacy framework, defining AI literacy as 'the comprehensive expression of the knowledge and skills, processes and methods, emotional attitudes and values gradually formed in the process of students receiving AI education' (36). In organisational contexts, Cetindamar et al. (2024) conceptualise employees' AI literacy as 'a collection of technology, work, human-

machine, and learning capabilities' enabling active participation in AI design and utilisation (11).

The theoretical frameworks underpinning AI literacy conceptualisation reflect diverse disciplinary orientations. Long and Magerko's (2020) framework identifies 17 competencies organised around five guiding questions: What is AI? What can AI do? How does AI work? How should AI be used? And how do people perceive AI? The AI4K12 Initiative's Five Big Ideas framework provides a pedagogically-oriented structure encompassing Perception, Representation and Reasoning, Learning, Natural Interaction, and Societal Impact (Touretzky et al. 2019). Ng et al. (2021) proposed a four-dimensional model based on Bloom's taxonomy: Know and Understand, Use and Apply, Evaluate and Create, and Ethics, subsequently refined to incorporate affective, behavioral, cognitive, and ethical (ABCE) dimensions (Ng et al. 2024a).

Through systematic analysis of existing frameworks, Almatrafi, Johri, and Lee (2024) identified six fundamental constructs underlying AI literacy: Recognize, Know and Understand, Use and Apply, Evaluate, Create, and Navigate Ethically. Despite variations in conceptualisation and operationalisation, these frameworks place convergent emphasis on both technical competencies and ethical considerations, while highlighting the relative priorities accorded to different dimensions across educational and professional contexts.

2.3. Distinguishing AI literacy from related constructs

Establishing clear boundaries between AI literacy and cognate constructs is essential for theoretical precision and empirical rigour. AI literacy intersects with but remains distinct from several related concepts that warrant clarification.

2.3.1. AI literacy versus AI education

Ng et al. (2024a) distinguish three related but distinct concepts: AI in education (adoption of AI technologies for learning purposes), AI education (specialised computer science education involving AI knowledge and skills), and AI literacy education (curricula, pedagogies, and assessment approaches to foster broad AI competencies). AI literacy education distinguishes itself from technical expertise in AI development by focusing on the development of fundamental competencies that enable informed engagement with AI systems. Of the three concepts, AI literacy research encompasses AI literacy education rather than AI in education and AI education.

2.3.2. AI literacy versus algorithmic literacy

Cox (2024) differentiates between algorithmic literacy and AI literacy: the former addresses understanding of algorithmic systems broadly conceived, whereas the latter specifically concerns machine learning-based systems characterised by data-driven learning and adaptive behaviour. This distinction is significant given that not all algorithmic systems involve AI, while AI systems increasingly extend beyond traditional algorithmic logic.

2.3.3. AI literacy versus GenAI literacy

The emergence of LLMs has prompted recognition of GenAI literacy as a potentially distinct construct. Cox (2024) proposes GenAI literacy encompassing competencies specific to generative systems, including prompt engineering, output evaluation, and understanding of model limitations and biases. Whether GenAI literacy constitutes a subset of AI literacy or represents an emergent construct requiring separate theoretical treatment remains an open question warranting empirical investigation.

2.3.4. AI literacy versus AI readiness

AI readiness typically refers to organisational or individual preparedness for AI adoption, encompassing attitudes, infrastructure, technical capabilities, and implementation capacity (Jöhnk, Weißert, and Wyrski 2021; Ramazanoglu and Akın 2025). While AI literacy contributes to AI readiness, the latter construct extends beyond competency development to encompass motivational, attitudinal, and contextual factors influencing AI engagement.

2.4. Critical appraisal of existing research syntheses

A growing body of research syntheses has attempted to consolidate knowledge in the AI literacy domain. Critical examination of these syntheses reveals both substantive contributions and significant limitations that collectively justify the present integrative review.

Seven bibliometric analyses have mapped the structural features of AI literacy research. Tenório et al. (2023) examined publication trends, collaboration networks, and thematic evolution prior to 2022, identifying leading contributors and publication venues but preceding the substantial research expansion following the public release of ChatGPT. Berigel and Şilbir (2024) extended bibliometric coverage to 2024, providing insights into publication trends, citation patterns, and structural development of the field. I-kitisiri, Hoonso-pon, and Puriwat (2025) used 436 Scopus-indexed

journal articles (2019–2025) to map publication trends, key contributors, journals, and thematic structures of AI literacy research. The findings show exponential growth in publications, peaking in 2024, and identify core journals, influential authors, and three main thematic clusters centered on educational technology and AI, curriculum and generative AI, and human/demographic dimensions. Kumar (2025) analysed 902 Scopus-indexed publications on AI Literacy (2021–early 2025), revealing a sharp rise in research output since 2023 driven by the emergence of GenAI. The identified key thematic clusters around ethics, education levels, GenAI, and digital literacies, while also highlighting gaps in methodological diversity, interdisciplinary participation, and global representation.

Two other bibliometric analyses specifically focused on education field. Yang et al. (2025) conducted a bibliometric analysis of 335 articles AI literacy education articles (2014–2024) using CiteSpace. The study identifies four developmental trajectories and nine key thematic clusters, including data literacy, machine learning, computational thinking, and technology acceptance, highlighting the interdisciplinary and expanding nature of AI literacy education research. Another bibliometric analysis conducted by Shoustikova et al. (2026) identified 154 articles published up to February 2025 in the area of AI literacy in educational field. Using VOSviewer, the analysis described publication trends, disciplinary focus, geographic distribution, and thematic patterns.

While these bibliometric studies offer valuable macro-level insights into publication trends, citation patterns, and thematic structures, they do not engage deeply with the substantive content, conceptual nuances, and contextual interpretation of findings.

Apart from bibliometric studies, a number of literature syntheses adopt approaches of scoping or systematic reviews, which have synthesised AI literacy research by focusing particular educational levels or specific aspect(s).

Su, Ng, and Chu (2023) examined AI literacy in early childhood education, identifying challenges including educator knowledge gaps, curricular shortages, and limited pedagogical guidelines for young learners. Reviewing AI literacy education at secondary education, Ng et al. (2024a) examined teaching tools, learning contents, and assessment methods of AI literacy education in secondary setting and identified collaborative project-based learning as the predominant pedagogical approach. Covering K-12 settings, Casal-Otero et al.'s (2023) review focused on learning experience and theoretical perspective of AI literacy research and also identified important gaps of these studies, including lacking

learning outcome assessment and insufficient attention to potential negative consequences of AI. Laupichler et al.'s (2022) scoping review of higher and adult education identified significant definitional inconsistencies and underscored the need to develop accessible educational content to better support non-expert learners. Sperling et al. (2024) conducted a scoping review of AI literacy in teacher education through the lens of Aristotelian knowledge forms (episteme, techne, phronesis) and found significant gaps in teachers' practical and ethical knowledge development.

Other specialised reviews have focused on different aspects of AI literacy research. Ng et al. (2021) conducted an early exploratory review of conceptualizations of AI literacy and proposed a four-dimensional AI literacy framework encompassing 'know and understand', 'use and apply', 'evaluate and create', and 'ethical issues'. This framework has subsequently informed many empirical studies and assessment instrument development. Almatrafi, Johri, and Lee (2024) systematically examined AI literacy conceptualizations and identified six key constructs, including *Recognize*, *Know and Understand*, *Use and Apply*, *Evaluate*, *Create*, and *Navigate Ethically*. This systematic review also examined implementation efforts, and assessment approaches of AI literacy. Evaluating 16 AI literacy instruments, Lintner (2024) reported that although the majority scales demonstrated adequate structural validity and internal consistency, few addressed reliability, content validity, or cross-cultural applicability. Table 1 summarises the existing AI literacy research syntheses.

While previous reviews have provided valuable overviews of AI literacy research, they have primarily focused on descriptive mapping of publication trends, conceptual definitions, or educational applications. Critical evaluation of these syntheses reveals four major limitations. First, purely bibliometric approaches, while valuable for mapping publication trends and citation networks, lack the analytical depth necessary for nuanced theoretical analysis and contextual understanding (Passas 2024). Second, traditional systematic reviews, while providing rigorous synthesis of empirical findings, do not adequately capture the intellectual communities and evolutionary trajectories characterising emerging research fields (Yang et al. 2023). Third, existing reviews typically focus on particular educational levels or specific aspects of AI literacy, precluding cross-contextual synthesis. Fourth, the rapid evolution of AI technologies, particularly GenAI, has outpaced the temporal coverage of existing syntheses, creating gaps in understanding contemporary developments.

These limitations collectively point to the need for an integrative approach that combines the capabilities of

Table 1. Summary of existing AI literacy research syntheses.

Study	Type	Limitations	No.	Year range
Tenório et al. (2023)	Bibliometric analysis	Lacking thematic synthesis; before public release of ChatGPT	244	1989–2021
Berigel and Şilbir (2024)	Bibliometric analysis	Lacking thematic synthesis	608	2016–2024
Kumar (2025)	Bibliometric analysis	Lacking thematic synthesis	902	2021–2025
I-kitisiri, Hoonsopon, and Puriwat (2025)	Bibliometric analysis	Lacking thematic synthesis		
Yang et al. (2025)	Bibliometric analysis	Lacking thematic synthesis; only focusing on AI literacy education	335	2014–2024
Shoustikova et al. (2026)	Bibliometric analysis	Lacking thematic synthesis; only focusing on AI literacy in educational field	154	2019–2025
Su, Ng, and Chu (2023)	Scoping review	Only focusing on AI literacy in early childhood education; before public release of ChatGPT	16	2016–2022
Ng et al. (2024a)	Thematic review	Only focusing on AI literacy in secondary education	50	2016–2022
Casal-Otero et al. (2023)	Systematic review	Only focusing on AI literacy in K-12 education	179	2004–2023
Laupichler et al. (2022)	Scoping review	Only focusing on AI literacy in higher and adult education; before public release of ChatGPT	30	2016–2022
Sperling et al. (2024)	Scoping review	Only focusing on AI literacy in teacher education	34	2000–2023
Ng et al. (2021)	Exploratory review	Only focusing on AI literacy conceptualisation; before public release of ChatGPT	49	2000–2020
Almatrafi, Johri, and Lee (2024)	Systematic review	Focusing on studies conceptualising and assessing AI literacy, and implementing AI literacy intervention	47	2019–2023
Lintner (2024)	Systematic review	Only focusing on scales assessing AI literacy	22	2021–2024

bibliometric analysis with the thematic depth of systematic synthesis. Such integration enables a more comprehensive examination of AI literacy research, addressing the methodological, scope, and temporal limitations characterising existing syntheses.

Combining bibliometric analysis with systematic synthesis, the present review responds to the above-mentioned limitations by moving beyond descriptive synthesis to offer a more integrative analysis of the field's conceptual, methodological, and contextual development. Specifically, this review synthesises how AI literacy has been theoretically framed across disciplines, examines the methodological orientations underpinning current evidence production, and interprets how sociocultural and educational contexts shape the conceptualisation and assessment of AI literacy, particularly in light of rapid GenAI-driven transformations. In doing so, the review not only consolidates existing findings but also identifies underlying tensions, methodological imbalances, and emerging trajectories that have remained underexplored in prior reviews. This contributes to a more theoretically and contextually nuanced understanding of the field and provides a foundation for future conceptual refinement and research development.

3. Methodology

3.1. Research design

This study employs an integrative review methodology combining bibliometric analysis with thematic synthesis. Unlike traditional systematic reviews focusing on empirical findings through standardised protocols, integrative reviews permit diverse research types and analytical approaches to generate in-depth

understanding of complex phenomena (Whittemore and Knafl 2005). This dual-method integration addresses inherent limitations of single-method syntheses. While bibliometric analysis enables macro-level mapping of publication trends and thematic evolution, thematic synthesis provides micro-level examination of study theme, methodological approaches, and research context (Marzi et al. 2025; Zupic and Čater 2015).

3.2. Search strategy

Systematic search was initially conducted in Scopus and Web of Science Core Collection at the end of January 2025, and subsequently updated in early April 2026 to capture the most recent publications in this rapidly evolving field. No start date restriction was applied to ensure a broad coverage of the field's development.

The search strategy was designed to capture the full spectrum of AI literacy and related constructs. Search terms were structured around two dimensions: (1) technology types, including foundational AI (AI, artificial intelligence), GenAI variants (GenAI, generative AI, GAI), and LLMs (LLM, large language model, ChatGPT, GPT); and (2) literacy constructs, including literacy, competency/competencies, capability/capabilities, and skill/skills. These dimensions were systematically combined to form search queries (Appendix A presents the complete search strings). Database-specific protocols were implemented: Web of Science employed Topic Search fields encompassing titles, abstracts, and keywords, while Scopus utilised TITLE-ABS-KEY fields. Supplementary searches included backward citation tracking of seminal papers and targeted search in Google Scholar to capture potentially missing literature.

We intentionally included broad terms ‘competency/competencies, capability/capabilities, and skill/skills’ in addition to literacy to maximise search sensitivity and ensure that studies employing these broad terms to describe AI literacy were not overlooked. However, to ensure conceptual alignment with AI literacy, all records retrieved using broad search terms (e.g. ‘AI skill’, ‘AI competency’, and ‘AI capability’) were subjected to a two-stage screening process. During the title and abstract screening, studies were retained only if AI literacy or closely related constructs (e.g. AI-related knowledge) were central to the study’s aim, framework, or measured outcomes. In the full-text review stage, inclusion was further restricted to studies in which these terms were explicitly used as synonymous with, or operationally defined in relation to, AI literacy. Studies that used these terms in unrelated technical or domain-specific contexts were excluded. For example, articles referring to ‘AI capability’ in terms of organisational or institutional infrastructure (e.g. enterprise-level AI capability or system deployment readiness), or ‘AI skill’ as machine performance or algorithmic capability rather than human learning, were not considered relevant. Only studies in which these terms were explicitly used to describe human-centred competencies or were conceptually aligned with AI literacy were included. This iterative screening process helped ensure that only studies substantively focused on AI literacy or its conceptual equivalents were included in the final synthesis.

3.3. Eligibility criteria

Studies were included if they: (1) focused primarily on AI literacy or related constructs (GenAI literacy, LLM literacy) as central research themes; (2) presented original contributions including theoretical frameworks, empirical findings, or assessment methodologies; (3) were published in peer-reviewed journals and conference proceedings; and (4) were published in English.

Studies were excluded if they mentioned AI literacy only peripherally, focused exclusively on organisational AI capabilities without human-centred literacy dimensions, or consisted of editorials, book chapters, and other non-peer-reviewed publications.

Conference proceedings were included to ensure comprehensive coverage of emerging and rapidly evolving research in AI literacy, where a substantial proportion of early-stage empirical and conceptual work is first disseminated through conferences before journal publication. Given the fast-moving nature of AI-related research, excluding conference papers could have led to the omission of relevant and timely contributions,

particularly those reflecting current educational practices and innovations.

To address concerns regarding quality, conference papers were subject to the same inclusion criteria as journal articles and were further screened for methodological and conceptual rigour during full-text review. Specifically, we included only conference papers that (1) demonstrated explicit relevance to AI literacy as a central construct rather than a peripheral topic; (2) provided clear research design and data or conceptual argumentation; (3) were sufficiently detailed to allow evaluation of methods and findings.

3.4. Study selection process

The selection process followed PRISMA 2020 guidelines (Page et al. 2021). The systematic search was initially conducted in January 2025 and updated in early April 2026 to capture the most recent publications in this rapidly evolving field. The combined database searches yielded 10,735 records (Scopus: 6,090; Web of Science: 4,645). Supplementary searches identified additional records through backward citation tracking of seminal papers ($n = 65$) and targeted search in Google Scholar ($n = 30$). Following the removal of 3,530 duplicates (285 identified manually and 3,245 through Covidence), two phases of independent screening (i.e. title and abstract; full-text) were performed between two authors from the research team using the inclusion and exclusion criteria. The title and abstract screening stage excluded 6,080 articles with Cohen’s κ inter-rater reliability between the two coders being .85. The full-text screening stage removed 387 articles with Cohen’s κ inter-rater reliability being .89 between the two coders. During coding, disagreements were resolved through discussion and third-reviewer consultation. The final corpus comprised 838 studies. The complete list of the 838 included studies is provided in Supplementary Material S1. The study selection process is presented in the PRISMA flow diagram (Figure 1).

3.5. Data analysis

3.5.1. Bibliometric analysis

Bibliometric analysis was performed using CiteSpace and VOSviewer which provided complementary insights. CiteSpace parameters included one-year time slicing across 2018–2026, g-index selection ($k = 25$), Link Retaining Factor (LRF) of 2.5, maximum links per node (L/N) of 10, look-back years (LBY) of 5, and Pathfinder network pruning for clarity of visualisation. VOSviewer employed association strength normalisation, full counting, with clustering resolution of 1.0

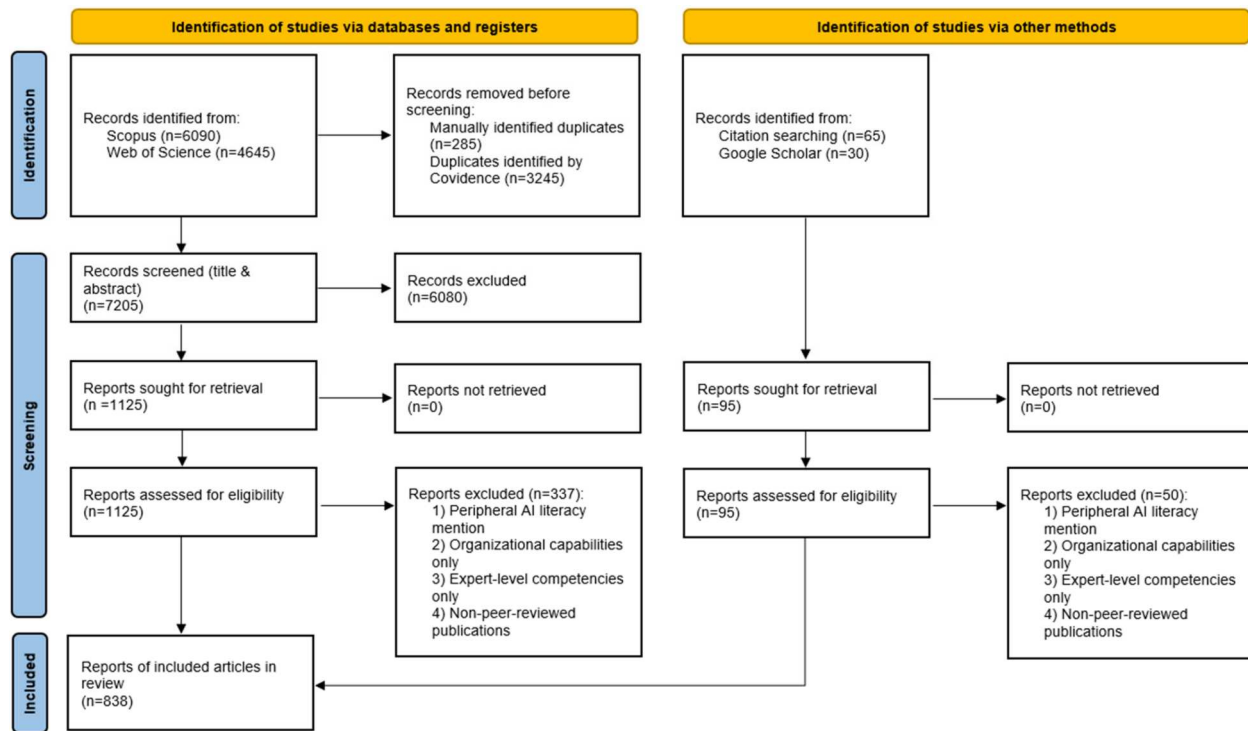


Figure 1. PRISMA diagram.

and minimum cluster size of 1; co-authorship and keyword co-occurrence analyses used minimum thresholds calibrated to the size of each network. Prior to analysis, author name disambiguation and keyword harmonisation were carried out included. Bibliometric analyses covered six elements: temporal trends, geographical distribution, institutional collaboration, publication venues, co-authorship networks, and keyword evolution.

3.5.2. Thematic synthesis

Thematic synthesis commenced with establishing a multidimensional coding framework (Braun and Clarke 2006). The framework encompassed seven dimensions: research theme, study type, methodological approach, target population, educational context, geographical region, and technological focus. Using this framework, a collaborative coding approach was adopted (Richards and Hemphill 2018). Two researchers collaboratively coded the entire dataset. Both researchers reviewed and analysed all data together, clarifying code interpretations and category boundaries throughout the process. Discrepancies were resolved through discussion and consensus. In cases where consensus could not be reached, a third coder was consulted to review the data and provide an independent judgment. This collaborative approach helped enhance consistency, reduce individual bias, and strengthen the credibility of the analysis.

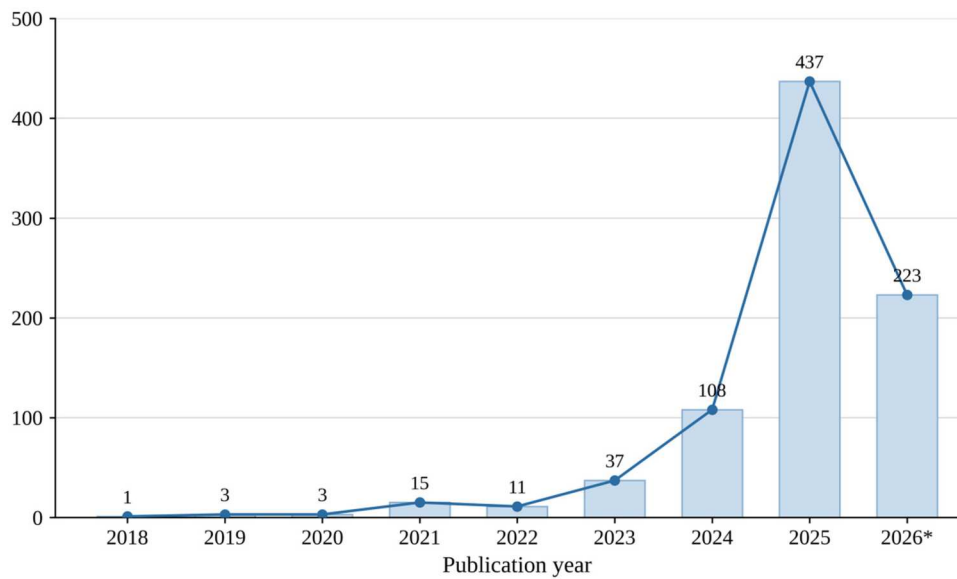
4. Results of RQ1 using bibliometric analysis

4.1. Temporal trends and publication growth

Figure 2 visualises the temporal distribution of AI literacy research in terms of annual publication frequency over the analysed period (2018–2026), which reveals a distinct pattern of substantial acceleration in scholarly outputs, of this emerging field.

The temporal distribution reveals three distinct phases in the evolution of AI literacy research:

- (1) a formative phase (2018–2020) characterised by conceptual groundwork and limited empirical investigation: The initial period of AI literacy as a formal research topic had limited production, with only one publication identified in 2018, followed by three publications in 2019 and 2020 respectively. These output figures reflect the nascent conceptualisation of AI literacy during this period, with foundational frameworks such as Long and Magerko's (2020) competency model only beginning to emerge.
- (2) an acceleration phase (2021–2022) marked by a sustained step-change in publication volume: A notable point of inflection occurred in 2021, when publication volume increased substantially to 15 articles, representing a 400% increase compared to the previous year. Output remained at a comparable



*2026 includes publications identified up to 3 April 2026, the date of the final search update.

Figure 2. Number of publications by year.

level in 2022 with 11 publications. This output pattern represents sustained efforts in developing AI literacy frameworks and interests in empirical AI literacy research.

- (3) an exponential expansion phase (2023–present) featuring the rapidly evolving nature of this emerging field: An unprecedented growth trajectory from 2023 onward was observed, with output more than tripling to 37 publications, marking a significant threshold in research productivity. The momentum continued in 2024, when publication volume nearly tripled again to 108 articles, representing a 192% increase over 2023. The most dramatic expansion occurred in 2025, when publications surged to 437, a 305% increase over 2024, with this single year alone accounting for 52.1% of the entire research corpus. The trajectory shows no signs of abating: the first three months of 2026 alone yielded 223 publications, an output that already exceeds the combined total of all publications between 2018 and 2023 ($n = 70$). Taken together, the 2024–2026 period accounts for 91.6% ($n = 768$) of the entire corpus, indicating that AI literacy has transitioned from a niche research interest into a substantive scholarly field in the span of barely three years.

The significant expansion of AI literacy is likely associated with the launch of ChatGPT on 30 November 2022 (Schulman et al. 2022), which generates widespread adoption of GenAI technologies in educational

and professional contexts, necessitating more robust understanding of literacy requirements to efficiently engage with various AI tools.

This exponential growth pattern aligns with typical bibliometric indicators of an emerging research field, characterised by initial conceptual development followed by rapid expansion as theoretical frameworks solidify and practical applications emerge (Chen 2006). The publication surge from 2023 onward coincides temporally with the emergence of GenAI technologies also corresponds with significant policy initiatives addressing AI literacy at institutional and governmental levels (Cardona, Rodríguez, and Ishmael 2023; Miao and Holmes 2021). The growth pattern seems to suggest a reciprocal relationship between research productivity, AI technological innovation, and societal recognition of AI literacy.

4.2. Geographical distribution of AI literacy output

Figure 3 presents the geographical distribution of AI literacy research, revealing pronounced regional concentration alongside considerable global dispersion. Analysis of first authors' affiliations across the 838 studies identifies 69 distinct countries and regions, indicating broad international engagement with AI literacy as a research domain. However, output is highly skewed toward a small number of dominant contributors: the United States leads with 170 studies (20.3%), followed by mainland China (131 studies, 15.6%), Hong Kong

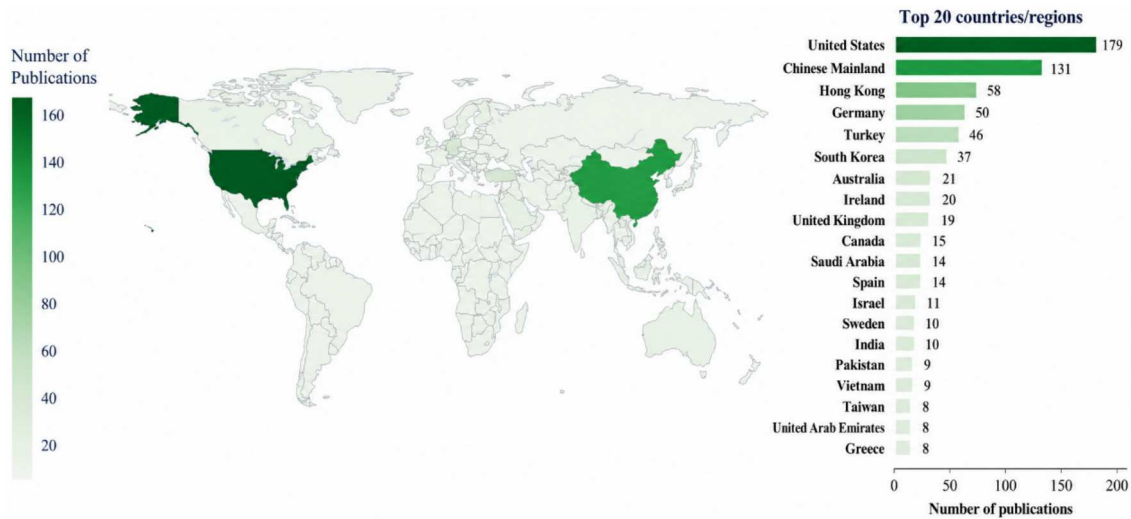


Figure 3. Distribution of AI literacy research output by regions.

(80 studies, 9.5%), and Germany (50 studies, 6.0%). These four regions collectively account for 51.4% of global output, underscoring their pivotal role in shaping the field's intellectual trajectory.

A discernible second tier comprises Turkey (46 studies, 5.5%), South Korea (37 studies, 4.4%), Australia (21 studies, 2.5%), Finland (20 studies, 2.4%), the United Kingdom (19 studies, 2.3%), and Canada (15 studies, 1.8%). Together with the top four, these ten regions account for 70.3% of the corpus, leaving the remaining 59 countries and regions to share less than 30% of total output.

A third tier of moderate contributors includes Saudi Arabia (14), Spain (14), Israel (11), Sweden (10), India (10), Pakistan (10), Vietnam (9), Taiwan (8), the United Arab Emirates (8), and Greece (8), alongside several European and Southeast Asian contributors, each producing between five and fourteen studies.

Notably, 41 of the 69 contributing regions (59.4%) produced five or fewer publications, and 23 produced two or fewer, indicating that despite the field's apparent global reach, sustained research programmes remain concentrated in a relatively narrow set of locations.

At the continental level, the distribution further reveals pronounced asymmetries. Asia-Pacific constitutes the largest contributing region (342 studies, 40.8%), driven primarily by an East Asian cluster comprising mainland China, Hong Kong, South Korea, Taiwan, Macau, and Japan (260 studies, 31.0%). When considered as an aggregated regional hub, Greater China (mainland China, Hong Kong, Taiwan, and Macau) alone contributes 220 studies (26.3%), reflecting concerted national and institutional investment in AI literacy research across Chinese-speaking academic communities. Europe follows with 243 studies (29.0%), and North America with 189 studies (22.6%).

In contrast, contributions from the Middle East (45 studies, 5.4%), Africa (10 studies, 1.2%), and Latin America (9 studies, 1.1%) remain markedly limited. South Asia is similarly underrepresented, with substantive output essentially confined to India and Pakistan (10 studies each). These disparities likely reflect uneven access to technological resources, research funding, and institutional infrastructure across regions, compounded by language and indexing biases inherent in major academic databases.

This skewed geographic distribution aligns with broader patterns in technology education research, in which regions with advanced technological infrastructure and substantial investment in AI education tend to dominate scholarly output (Crompton and Burke 2023; Song and Wang 2020; Wang et al. 2024). However, such concentration also signals a possible widening AI literacy divide: theoretical frameworks, assessment instruments, and pedagogical models are being developed predominantly within a narrow set of high-resource contexts, raising critical questions about their cultural assumptions, conceptual transferability, and applicability to learners in the Global South. Without deliberate efforts to broaden research participation through targeted funding mechanisms, cross-regional collaborations, and the co-development of culturally responsive frameworks, AI literacy research risks reproducing the very digital and educational inequities it might otherwise help address.

4.3. Institutional collaborations

Analysis of the institutional collaboration network (Figure 4) reveals a research andscape that remains sparsely connected overall but is beginning to

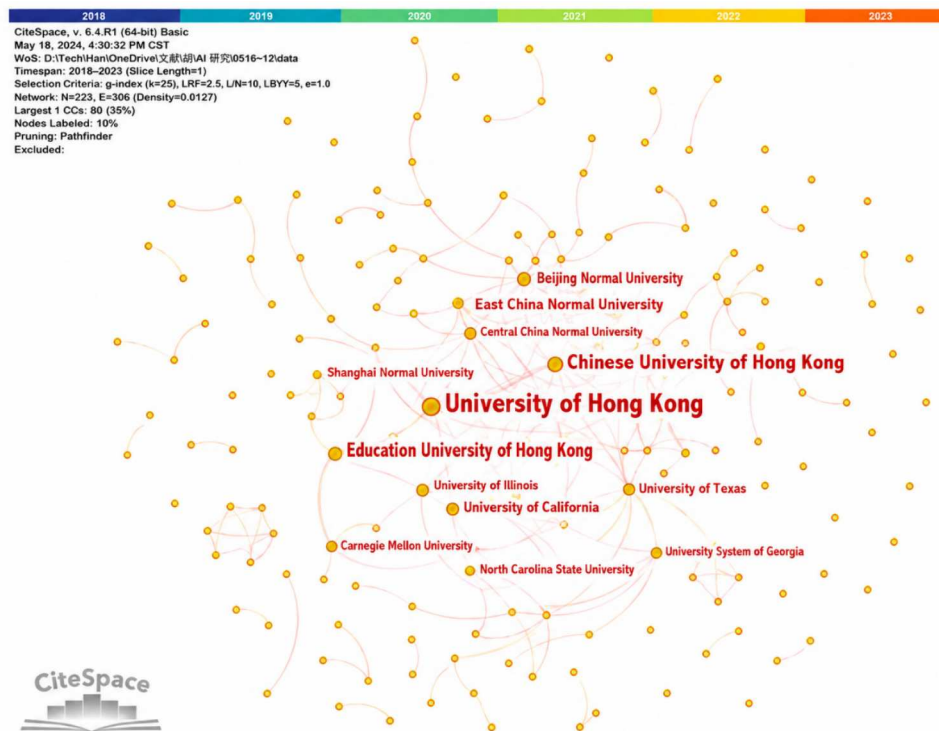


Figure 4. Institutional collaborations.

consolidate around several identifiable hubs. The network comprises 232 institutional nodes connected by 205 collaboration edges, yielding a low overall density of 0.0077 and indicating that most cross-institutional partnerships are still confined to small clusters or isolated dyads. At the same time, the largest connected component encompasses 80 nodes (approximately 34% of the network), suggesting that a substantial collaborative backbone has begun to form alongside the long tail of peripheral institutions.

Within this connected core, two regional clusters are clearly distinguishable. The first and most prominent is centred on Hong Kong-based institutions: the University of Hong Kong, the Education University of Hong Kong, and the Chinese University of Hong Kong emerge as the network's most central nodes, with the University of Hong Kong occupying the structurally most prominent position. These institutions anchor a collaborative cluster that extends primarily to mainland Chinese universities, particularly a sub-cluster of teacher education universities (Beijing Normal University, East China Normal University, Central China Normal University, and Shanghai Normal University). This pattern suggests concentrated efforts to integrate AI literacy into teacher education programmes within Chinese-speaking academic communities, and is consistent with the substantial publication output identified in Section 4.2 for Greater China.

A second, less consolidated cluster has emerged around US institutions, including the University of California, the University of Texas, the University of Illinois, Carnegie Mellon University, North Carolina State University, and the University System of Georgia. This cluster reflects the leading position of the United States in overall publication volume and indicates that AI literacy research in the US is distributed across both computer-science-strong institutions (e.g. Carnegie Mellon, University of Illinois) and large public university systems. However, direct collaboration links between the US cluster and the East Asian hub appear limited within the visualisation, suggesting that trans-Pacific partnerships in AI literacy research remain underdeveloped despite the prominence of both regions in the publication record.

Beyond these two hubs, connections to institutions in Europe, Southeast Asia, the Middle East, and other regions remain sparse and largely peripheral, reflecting untapped potential for cross-regional knowledge exchange.

The persistent presence of isolated nodes across the periphery of the network indicates that a considerable share of AI literacy research is still produced by institutions working without sustained collaborative ties. The centralisation around a small number of hub institutions, while facilitating concentrated knowledge production and rapid framework development, also

highlights structural vulnerabilities in knowledge diffusion. Strengthening collaborative infrastructure, particularly trans-regional partnerships that bridge the East Asian and North American hubs with underrepresented institutions in the Global South, remains a critical priority for the field's continued maturation and for ensuring the diverse perspectives that sustainable field development requires (Yang et al. 2023).

4.4. Publication venues analysis

Publication venues were analysed separately for journal articles and conference proceedings. Of the 838 studies in the corpus, 675 (80.5%) appeared in peer-reviewed journals and 163 (19.5%) in conference proceedings (a ratio of four to one), indicating that journals are the dominant channel for AI literacy research dissemination.

4.4.1. Journal article analysis

The 675 journal articles are distributed across 303 outlets, indicating broad but highly skewed dissemination. Two journals lead the field by a clear margin: *Computers and Education: Artificial Intelligence* (35 articles, 5.2%) and *Education and Information Technologies* (34 articles, 5.0%), together accounting for 10.2% of journal output. Both align thematically with the intersection of educational practice and AI innovation, and the rapid rise of *Computers and Education: Artificial Intelligence*, a relatively new outlet, signals the consolidation of dedicated publication infrastructure for the field (Figure 5).

A second tier of eight journals each publishing 10–15 articles (15.0% of journal output) spans education-focused outlets (*Frontiers in Education*, *Interactive Learning Environments*, *Education Sciences*, *European Journal of Education*, *Computers and Education Open*) alongside human-computer interaction- and psychology-oriented venues (*International Journal of Human-Computer Interaction*, *Behavioral Sciences*, *Frontiers in Psychology*), reflecting the field's expansion beyond educational technology into adjacent disciplinary communities.

Within the third tier (13 journals, 5–9 articles each), two discipline-specific clusters become visible: a health professions cluster (e.g. *BMC Medical Education*, *Nurse Education in Practice*, *Nurse Education Today*) and a library and information science cluster (e.g. *Journal of Academic Librarianship*, *the International Federation of Library Associations and Institutions (IFLA) Journal*, *Library Trends*), suggesting that AI literacy is increasingly treated as a domain-specific competency rather than a generic educational concern.

Beyond these tiers, the remaining 280 journals each contributed four or fewer articles, with 197 (65.0% of outlets) publishing only a single article on AI literacy. This long tail extends across diverse fields such as marketing education, language teaching, chemistry education, and journalism studies, attesting to the field's striking interdisciplinary reach.

4.4.2. Conference proceedings analysis

Conference venues typically allow faster dissemination than traditional journal cycles, which suits AI literacy research as it responds to rapid technological change. As shown in Figure 6, the 163 conference papers are distributed across 73 venues, with three communities anchoring the field. Human-computer interaction is the most prominent, led by the *ACM Conference on Human Factors in Computing Systems* (19 papers, 11.7%) and the *ACM Interaction Design and Children Conference* (11 papers, 6.7%), reflecting strong human-computer interaction dimensions of AI literacy research, particularly for younger learners.

AI-in-education forms a second cluster anchored by the *International Conference on Artificial Intelligence in Education* (17 papers, 10.4%). A computer-science-education cluster comprises the *ACM Technical Symposium on Computer Science Education* (8 papers, 4.9%), the *ACM Conference on Innovation and Technology in Computer Science Education* (6 papers, 3.7%), and the *Koli Calling International Conference on Computing Education Research* (4 papers), underscoring the field's continuing connection to computing pedagogy.

Beyond these anchors, 65 further venues each contributed four or fewer papers, with 49 (30.1% of all conference papers) producing a single contribution. The dispersion spans fairness and accountability (e.g. the *ACM Conference on Fairness, Accountability, and Transparency*), mainstream AI (e.g. the *Association for the Advancement of Artificial Intelligence Conference on Artificial Intelligence*), robotics education (e.g. the *International Conference on Robotics in Education*), and information science (e.g. the *Proceedings of the Association for Information Science and Technology*), mirroring the broad interdisciplinary footprint observed for journal venues.

4.4.3. Co-authorship network analysis

The co-authorship network (Figure 7) reveals an evolving collaborative landscape comprising three thematically and regionally distinct clusters connected by a small intermediate group. Temporal progression is represented through the colour spectrum, with darker hues (2023) indicating earlier collaborations and lighter hues (2025–2026) marking emergent contributors.

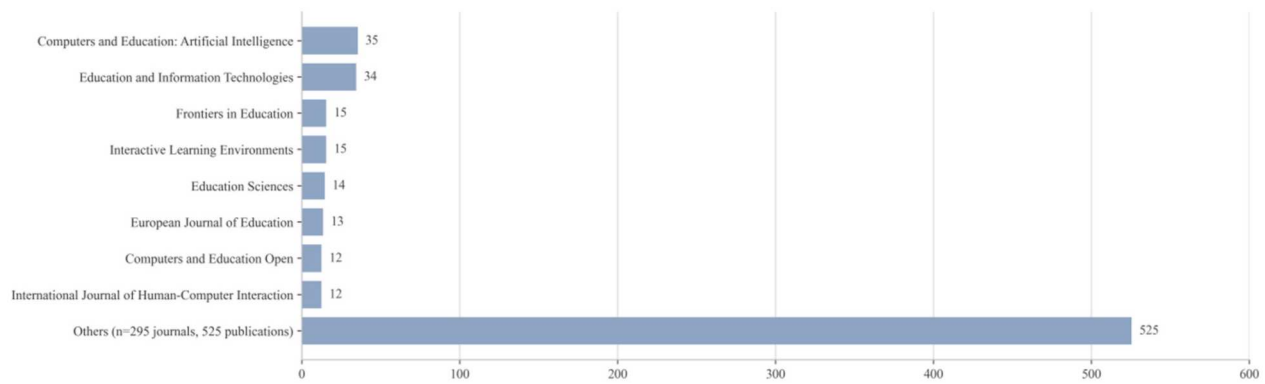


Figure 5. Major journals publishing AI literacy research.

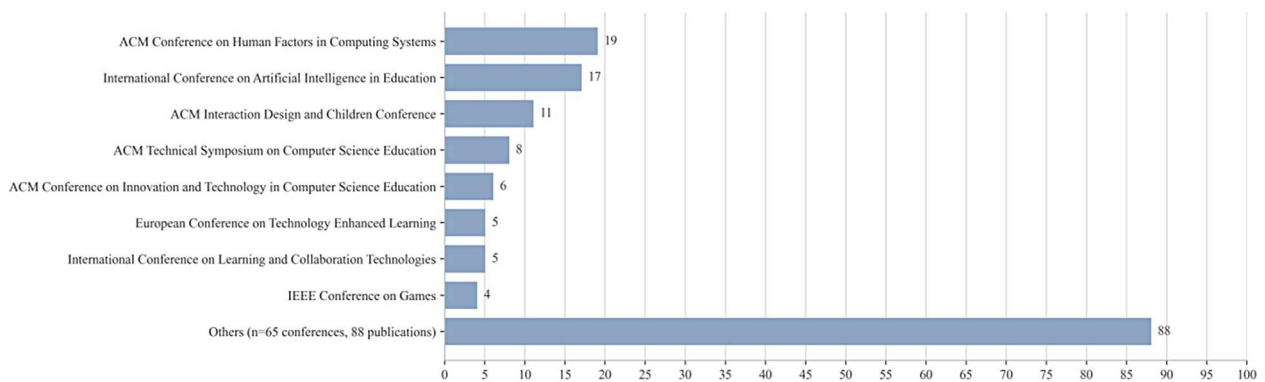


Figure 6. Major venues of conference proceedings for AI literacy research.

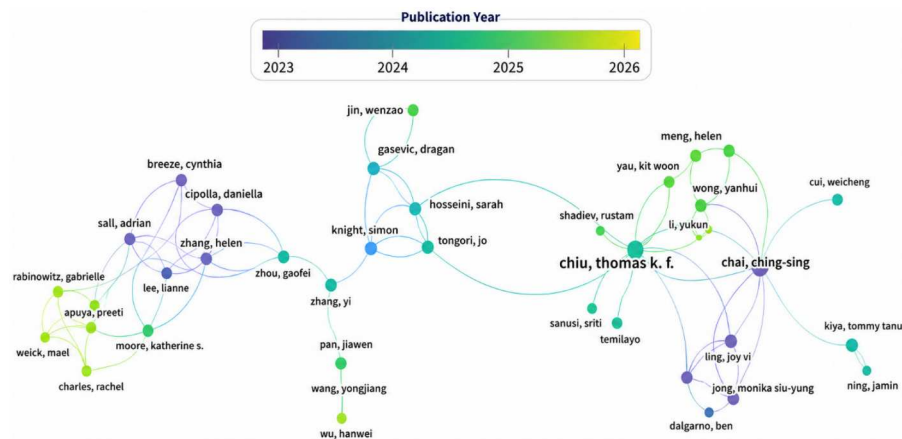


Figure 7. Co-authorship networks.

The most prominent cluster, located on the right of the network, is centred on Hong Kong-based scholars working on AI literacy curricula, instrument development, and teacher competency development. Thomas K. F. Chiu and Ching-Sing Chai (both at the Chinese University of Hong Kong) serve as the principal hubs, surrounded by a dense set of collaborations involving Yun Dai, Morris Siu-Yung Jong, Pei-Yi Lin, Helen

Meng, Irwin King, King Woon Yau, and Yeung Yam (Chai et al. 2021; Chiu and Chai 2020). A more recently formed satellite at the network's far right, anchored by Tommy Tanu Wijaya, reflects post-2024 expansion into latent-profile analyses of teachers' and students' AI literacy in Chinese contexts (Wijaya et al. 2024).

A second cluster on the left brings together two related but institutionally distinct groups working on

K-12 and informal AI literacy education in the United States. The MIT Media Lab group, including scholars Irene Lee, Safinah Ali, Cynthia Breazeal, Daniella DiPaola, and Helen Zhang, has produced influential work on AI literacy curricula and educational technologies for middle-school learners (Lee et al. 2021; Williams et al. 2024; Zhang et al. 2023). Adjacent to this group, a satellite of researchers at the American Museum of Natural History (Gabrielle Rabinowitz, Mark Weckel, Preeti Gupta, Rachel Chaffee) collaborates with MIT-affiliated researchers (Katherine S. Moore, Safinah Ali, Irene Lee) on machine-learning-integrated science curricula for high-school youth in informal learning settings (Rabinowitz et al. 2025).

Between these two regional hubs, a smaller intermediate group performs a partial bridging role. It includes a Monash University Centre for Learning Analytics cluster (Yueqiao Jin, Dragan Gašević, and colleagues) developing GenAI literacy assessment tools (Jin et al. 2025), together with Sarah Howard (now at Leeds, previously Wollongong) and Jo Tondeur (Vrije Universiteit Brussel), who work on teacher technology integration and have recently co-authored with Chiu on teacher AI literacy (Chiu et al. 2026). Their position at the centre of the network reflects emerging trans-regional collaboration around teacher AI competency and assessment.

Overall, the network exhibits a ‘hub-and-bridge’ structure: two dense regional clusters anchored by senior scholars, linked through a smaller intermediate cluster of Australian-European researchers. While this structure supports concentrated knowledge production within each hub, the relatively sparse cross-cluster ties indicate potential for stronger trans-regional collaboration, particularly with researchers in underrepresented regions, to ensure diverse perspectives and sustainable field development (Yang et al. 2023).

4.5. Temporal evolution of AI literacy research themes

The CiteSpace time-zone visualisation (Figure 8) maps the emergence of dominant keywords from 2018 to 2026, revealing five developmental phases in the field’s intellectual trajectory.

4.5.1. Foundational phase (2018–2019)

Early research established the conceptual scaffolding of AI literacy. In 2018, dominant keywords, such as *AI*, *AI education*, *students*, *curriculum*, *K-12 education*, indicate that AI literacy initially emerged through adaptations of computer science and curriculum-design traditions to AI-related content (Ho et al. 2019). The decisive moment came in 2019, when AI literacy itself

appeared as a distinct keyword (frequency = 637, the highest in the entire corpus), alongside literacy, education, and adoption, marking the field’s formal consolidation as a research construct distinguishable from earlier, more diffuse work on AI education (Long and Magerko 2020).

4.5.2. Framework expansion and ethical broadening (2020–2022)

This phase witnessed the diversification of AI literacy beyond purely technical understanding. In 2020, *AI ethics* (frequency = 44), *machine learning*, *AI anxiety*, and *AI readiness* emerged, signalling growing attention to ethical, affective, and preparatory dimensions of AI engagement. In 2021, *digital literacy*, *data literacy*, *computational thinking*, and *AI in education* situated AI literacy within the broader ecosystem of cognate literacies, while *acceptance*, *attitudes*, and *behavioural intention* foreshadowed the technology-acceptance research stream that would expand substantially in subsequent years. Contextual expansion appeared in 2022 with *higher education* (frequency = 71) and *early childhood education*, indicating the field’s extension across the full educational spectrum.

4.5.3. Genai emergence and theoretical consolidation (2023)

The year 2023 marks a decisive thematic shift. *Generative AI* emerged as the second-most-frequent keyword in the entire corpus (frequency = 155), coinciding with the post-ChatGPT diffusion of large language models into educational and professional contexts. Concurrent keywords including *teacher education*, *self-efficacy*, *pedagogical content knowledge*, *frameworks*, *scale*, and *validation*, indicate two parallel developments: a rapid pivot toward GenAI-specific competencies, and consolidation of the theoretical and measurement infrastructure underpinning AI literacy more broadly. The simultaneous prominence of teacher education and pedagogical content knowledge signals growing recognition that educators themselves require systematic AI literacy preparation.

4.5.4. Empirical and psychometric maturation (2024)

By 2024, the keyword landscape reflects substantial methodological sophistication. *Technology acceptance model*, *user acceptance*, *structural equation modelling*, and *scale development* dominate, indicating consolidation of psychometric and complex modelling research designs. Concurrent emergence of *pre-service teachers*, *professional development*, *self-regulated learning*, *motivation*, and *information literacy* signals the field’s expansion into educator preparation and learner-process



The most recent phase exhibits two distinctive

The temporal analysis reveals four patterns of

Persistent gaps warrant attention. Critical perspectives

Table 2. Thematic synthesis coding framework.

Dimension	Categories	Description
Research theme	e.g. relations between AI literacy and other constructs; educational strategies and curriculum design; development of theoretical and conceptual frameworks; instrument development and validation	Primary conceptual focus of the study
Methodological approach	empirical study; theoretical/conceptual study; review (e.g. systematic review, scoping review)	Fundamental methodological approach
Research design	quantitative methods; qualitative methods; mixed-methods	Primary research design for data collection and analysis
Target population	e.g. university students; K-12 students; teachers/educators; general populace	Primary population studied or addressed
Educational context	e.g. higher education; K-12 education; early childhood education; adult education	Educational setting or level
Research location	e.g. mainland China; USA; Hong Kong; multi-country; Germany.	The country or region where the research was conducted
Technological focus	e.g. GenAI; AI chatbots/ai assistants; AI-powered educational tools; robotics	Primary AI technology addressed or employed

insights into this emerging field. Through detailed coding of the 838 included studies using a multidimensional coding framework (Table 2), we identified patterns in research themes, methodological approaches, and target population, and educational contexts that characterise the evolving landscape of AI literacy research.

5.1. Research themes

To systematically understand the intellectual landscape of AI literacy research, we categorised the 838 included studies into six distinct categories by their themes. While acknowledging that many studies address multiple themes concurrently, each publication was assigned to a single primary category based on its predominant research focus and contribution, consistent with methodological approaches in bibliometric and systematic synthesis studies (Hallinger and Chatpinyakoo 2019). This analytical decision reduces the thematic complexity inherent in interdisciplinary research at the same time enables clearer identification of dominant research focus. The thematic categorisation process involved iterative coding and validation by multiple researchers to ensure reliability, with disagreements resolved through discussion (O'Connor and Joffe 2020).

As presented in Table 3, two thematic categories jointly dominate the field: 'educational strategies and

curriculum design' (31.6%) and 'relations between AI literacy and other constructs' (30.5%). 'Development of theoretical and conceptual frameworks' accounts for 15.5%, followed by 'extent of AI literacy' (9.3%), 'instrument development and validation' (8.1%), and 'tools and systems development' (4.9%).

5.1.1. The 'educational strategies and curriculum design' theme

Studies in the 'educational strategies and curriculum design' category (31.6%) focus on pedagogical approaches, instructional frameworks, and curriculum implementation across educational levels. Two broad orientations are discernible. The first develops bespoke pedagogies tailored to AI literacy, exemplified by Arn and Huang's (2024) storybook approach for kindergarten children, Dai's (2024) dual-contrast pedagogy using human-AI comparison for upper-elementary learners, Zhang et al.'s (2023) DAILY workshop integrating technical concepts, ethical implications, and career awareness for middle-school students, Ng et al.'s (2024c) project-based AI-maker activities for Hong Kong secondary students, and Bellas et al.'s (2024) robotics-based Robobo Project applicable across age groups.

The second orientation embeds AI literacy into existing curricula, as in Ngo and Hastie's (2025) integration of AI literacy into an English for Academic Purposes module, Naamati-Schneider and Alt's (2024)

Table 3. Distribution of research themes in AI literacy literature.

Research theme	Frequency	Percentage	Primary focus
Educational strategies and curriculum	265	31.60%	Pedagogical approaches, curriculum design, instructional strategies for AI literacy education or professional development
Relations between AI literacy and other constructs	256	30.50%	Relations between AI literacy and psychological, cognitive, educational constructs, or technology acceptance
Development of theoretical and conceptual frameworks	130	15.50%	Framework development, conceptualisation, and operationalisation of AI literacy
Extent of AI literacy	78	9.30%	Assessment and evaluation of proficiency of AI literacy
Instrument development and validation	68	8.10%	Development and validation of standardised assessment tools to measure AI literacy
Tools and systems development	41	4.90%	Design and evaluation processes of educational platforms, online systems, and technological tools for AI literacy education

combination of problem-based learning and ChatGPT activities, and DeVasto and Palmer's (2024) incorporation of AI topics into business communication instruction. A particularly active strand embeds GenAI literacy into discipline-specific instruction across business and marketing classrooms (Beninger et al. 2025; MacArthur et al. 2025), academic writing in EFL contexts (Woo et al. 2026), and language teacher education (Karaduman 2025), suggesting that the integrative orientation is increasingly favoured as GenAI tools become routinely available in subject-area instruction.

Of diverse educational strategies and curriculum design, the majority of them target formal learning, with a smaller subset addressing informal settings, including embodied and creative public engagement (Zhang, Cocking, et al. 2024), family-centred AI literacy (Druga, Christoph, and Ko 2022), and teacher self-study communities (Yang and Banks 2024). Intervention studies emphasise constructivist and experiential principles, moving beyond conventional instruction towards project-based, inquiry-driven, and collaborative methodologies.

However, the body of intervention research warrants two interpretive cautions. The pedagogical pluralism evident reflects exploratory diversification rather than evidence-based consolidation: few interventions have been replicated across contexts or compared with a valid control group, leaving questions about relative effectiveness largely unresolved. Learning outcomes are typically assessed immediately after the intervention using study-specific measures, with longer-term retention and transfer of AI literacy competencies rarely tracked, a limitation that constrains both theoretical generalisation and practical translation of findings into curriculum policy.

5.1.2. The 'relations between AI literacy and other constructs' theme

Research on 'relations between AI literacy and other constructs' (30.5%), the second-largest theme in the corpus, investigates how AI literacy interacts with established psychological, cognitive, educational, and behavioural variables. Psychological connections examined include self-efficacy (Sun et al. 2023), innovation mindset (El-Sayed et al. 2025), psychological well-being (Fu 2025), and emotional engagement (Yang et al. 2024). Cognitive constructs include creativity (Zhang, Rice, and Wang 2026), higher-order thinking (Lu et al. 2025), and cognitive absorption (Celik 2023). Educational variables include learning engagement and peer interaction (Lu et al. 2025) and academic performance (Singh, Vasishta, and Singla 2024), while professional outcomes include work performance among

medical librarians (Mughari, Rafique, and Ali 2024). Studies in this category typically employ correlation, regression, path analysis, or structural equation modelling to estimate bivariate or multivariate relations.

A substantial parallel stream examines how AI literacy relates to users' acceptance and adoption of AI technologies, drawing on a range of behavioural-intention models including the Perceived Value (Al-Abdullatif and Alsubaie 2024), the Theory of Planned Behaviour (Wang, Wang et al. 2025), the Unified Theory of Acceptance and Use of Technology (Jang 2024), and the Technology Acceptance Model (Schiavo, Businaro, and Zancanaro 2024).

Beyond these variable-centred approaches, person-centred techniques, particularly latent profile and cluster analyses, identify distinct subgroups defined by different configurations of AI literacy and adjacent competencies. Profile-based studies span mathematics teachers (Wijaya et al. 2024), pre-service second-language teachers (Wu, Wang, and Lalli 2026), preschool teachers (Xiong et al. 2025), nursing students (Zhu et al. 2026), university learners (Medina-Gual and Parejo 2026; Zhou, Curle, and Zou 2025), and Chinese college students (Zhang and Kang 2026). The proliferation of profile-based work signals a recognition that aggregate-level associations may obscure substantial heterogeneity in how AI literacy is configured across individuals, and that the identification of population-specific patterns often carries greater pedagogical relevance and practical utility than average effects.

Despite the analytical richness of this category, the relational evidence base warrants careful interpretation. The overwhelming majority of studies are cross-sectional, which restricts causal inference: whether AI literacy drives outcomes such as engagement, well-being, or performance, whether prior engagement and performance shape AI literacy development, or whether both are jointly determined by unmeasured antecedents such as prior digital exposure or socio-economic background. Relational studies also rely overwhelmingly on self-report instruments, raising recurrent questions about the discriminant validity of AI literacy relative to adjacent constructs such as digital literacy and technology self-efficacy. Longitudinal and quasi-experimental designs, together with closer attention to construct distinctiveness, will be needed to convert the proliferation of associations into more defensible claims.

5.1.3. The 'development of theoretical and conceptual frameworks' theme

The 'development of theoretical and conceptual frameworks' theme (15.5%) focuses on establishing foundational conceptualizations of AI literacy and developing

frameworks for understanding its dimensions and components. Long and Magerko's (2020) seminal work, identifying 17 core competencies across five domains, established a foundation that has informed much subsequent work. Beyond this domain-based approach, two further orientations are visible. One categorises AI literacy through discrete component models, such as Ng et al.'s (2021) four-aspect model (Know and Understand, Use and Apply, Evaluate and Create, Ethical Issues). The other tailors frameworks to specific educational or professional settings, including Luo et al.'s (2024) SIACC framework for early childhood, Aleman and Martinez's (2024) ED-AI Lit framework for K-12 education, Cetindamar et al.'s (2024) framework for professional contexts, and Lorenz and Romeike's (2023) AI-PACK model extending TPACK to AI competencies. Theoretical work has responded quickly to technological evolution: Cox (2024) differentiated algorithmic literacy, AI literacy, and responsible GenAI literacy, while Mishra, Warr, and Islam (2023) adapted TPACK for the GenAI era. Critical perspectives have also begun to enter the literature, with Heyder and Posegga (2021) foregrounding sociocultural dimensions and Picasso et al. (2024) advancing a data-justice approach.

The theoretical landscape is largely characterised by framework proliferation than by framework consolidation. New conceptualizations are introduced more rapidly than existing ones are empirically tested, comparatively evaluated, or iteratively consolidated, and the relations among frameworks, whether complementary, overlapping, or competing, are rarely compared and articulated. The modest share of the corpus occupied by this category (15.5%), set against the rapid growth of empirical work that draws on these frameworks, suggesting that the lack of theoretical synthesis may pose a major obstacle to the field's further advancement and maturation.

5.1.4. The 'extent of AI literacy' theme

Studies in the 'extent of AI literacy' category (9.3%) examine current levels of AI literacy across diverse populations, including students (medical students: Subaveerapandian et al. 2024; K-12 students: Kasinidou, Kleanthous, and Otterbacher 2024), teachers (primary and secondary teachers: Zhao, Wu, and Luo 2022; higher-education educators: Lérias, Guerra, and Ferreira 2024), and professionals (medical physics experts: Diaz et al. 2021). Most studies focus on a single country or region, with comparatively few cross-national comparisons; where conducted, such comparisons reveal substantial geographic variation (Mansoor et al. 2024; O'Dea et al. 2026). A discernible strand within this

category profiles teachers' AI and GenAI literacy in disciplinary and linguistic contexts, particularly within Chinese EFL and second-language teacher populations (Karaduman 2025; Pan and Wang 2025), and maps disciplinary differences in students' GenAI literacy (Zhou, Curle, and Zou 2025).

Only a small number of studies in this category employ qualitative methodologies, such as open-ended survey responses, semi-structured interviews, or focus groups, to explore learners' existing understandings and conceptions of AI literacy (Heeg and Avraamidou 2025; Mertala and Fagerlund 2024; Velander et al. 2024). These qualitative approaches allow for more nuanced insights into how learners interpret and make sense of AI-related concepts in their own words, providing more actionable evidence for targeted interventions. In contrast, the majority of studies in this area adopt a quantitative research design, typically relying on closed-ended questionnaires to measure AI literacy in a structured and standardised manner. While this approach enables comparison across larger samples and supports statistical analysis, it may also limit the depth of understanding regarding learners' underlying reasoning, perceptions, and contextual interpretations of AI literacy.

A persistent methodological concern across this category is the near-universal reliance on self-reported literacy rather than performance-based evidence. Because self-assessment of competence is known to correlate imperfectly with demonstrated proficiency, inferences about absolute proficiency levels should be treated with caution. The field would benefit from broader incorporation of behavioural and task-based assessments alongside self-report scales.

5.1.5. The 'instrument development and validation' theme

Research in the 'instrument development and validation' category (8.1%) focuses on creating and validating tools to measure AI literacy across diverse contexts. These studies predominantly employ rigorous psychometric methods, with exploratory and confirmatory factor analysis as the primary analytical approaches (Ma and Chen 2024; Wang and Chuang 2024). A four-dimensional framework (Awareness, Usage, Evaluation, Ethics) originally proposed by Wang, Rau, et al. (2023) has gained substantial traction, with multiple instruments adopting this structure (Hobeika et al. 2024; Ma and Chen 2024). Alternative frameworks include Ng et al.'s (2024b) ABCE approach (Affective, Behavioural, Cognitive, Ethical) and Zhang, Perry, and Lee's (2025) content-focused domains. Instruments have been developed for a wide range of target populations, including

early childhood (Su 2024), K-12 students (Chen et al. 2024; Zhang, Perry, and Lee 2025), university students (Hornberger, Bewersdorff, and Nerdel 2023), teachers (Li and Liang 2025; Ramazanoglu and Akin 2025), and the general public (Koch et al. 2024; Yuan, Tsai, and Chen 2024), as well as for specialised contexts such as medical education (Karaca, Çalışkan, and Demir 2021) and foreign language instruction (Li and Liang 2025).

A distinct GenAI-specific sub-stream of instrument development has formed alongside the diffusion of large language models, including Lee and Park's (2024) ChatGPT Literacy Scale, Wang, Derakhshan et al.'s (2025) GenAI Literacy Scale designed for EFL teachers, Wu, Wang, and Lalli's (2026) GenAI Competence Scale with latent-profile identification for pre-service second-language teachers, Gümüş and Kara's (2025) Generative AI Literacy for Learning Scale, and Yan, Liu, et al.'s (2025) GenAI Literacy Scale linked to privacy-protection and information-verification behaviours.

This sub-stream reflects a growing recognition that GenAI competencies, prompt formulation, hallucination detection, evaluation of generated content, and responsible co-creation, which may warrant measurement separately from general AI literacy, though the theoretical relation between GenAI literacy and broader AI literacy remains under-specified.

Although many instruments demonstrate adequate structural validity and internal consistency, Lintner (2024) identified important limitations regarding content validity, cross-cultural applicability, and the reporting of reliability and measurement error, indicating clear directions for psychometric refinement. An imbalance between self-report and performance-based assessment also persists, with only three instruments in the corpus measuring actual performance (Hornberger, Bewersdorff, and Nerdel 2023; Wang and Chuang 2024; Zhang, Perry, and Lee 2025). The modest share of the corpus devoted to instrument development (8.1%), set against the rapid growth of relational research (30.5%) that relies on these instruments, points to a methodological imbalance.

5.1.6. The 'tools and systems development' theme

Studies in the 'tools and systems development' category (4.9%), which accounts the smallest category proportionally, focus on creating and evaluating educational technologies and platforms designed or adapted to develop AI literacy. Examples include Williams et al.'s (2024) Doodlebot educational robot for K-12 students, Chookaew et al.'s (2024) Smart Farming Artificial Intelligence of Things platform for computer-science university students, and Bilstrup et al.'s (2024) ml-machine.org

platform for school learners and teacher training. Despite the appearance of design-oriented work on age-appropriate AI learning technologies for children (Jia and Yu 2025), purpose-built tools for early childhood and informal learning settings remain notably scarce. Sustained investment in such tools, together with rigorous empirical evaluation, represents an important direction for advancing AI literacy across the full developmental spectrum.

5.2. Types of research and methodological approaches

Analysis of types of research and methodological approaches provides critical insights into how AI literacy is being investigated and what types of evidence substantiate knowledge claims in this field. Following established classification frameworks in educational research (Hallinger and Chatpinyakoo 2019), studies were categorised into three primary types: empirical, theoretical/conceptual, and review/meta-analysis, with empirical works further classified by methodological orientation (quantitative, qualitative, or mixed-methods).

As illustrated in Figure 9, empirical studies constitute approximately four-fifths of the research corpus (678 studies, 80.9%), with theoretical/conceptual works (86 studies, 10.3%) and reviews (74 studies, 8.8%) comprising the remainder. This distribution suggests that AI literacy research has progressed beyond initial conceptualisation to emphasise empirical investigation, indicative of a gradual maturing trend.

Among empirical studies, quantitative methodologies remain dominant (377 studies, 55.6%), followed by mixed-methods (179 studies, 26.4%) and qualitative approaches (122 studies, 18.0%).

5.2.1. Quantitative research and methodological approaches

Quantitative research in AI literacy exhibits considerable methodological diversity while maintaining a strong emphasis on measurement and association. Survey-based studies employing Likert-scale instruments represent the dominant approach within quantitative works. These studies predominantly employ structural equation modelling to examine relationships between AI literacy and psychological, cognitive, and educational constructs (e.g. self-efficacy, motivation, behavioural intention) (Al-Abdullatif and Alsubaie 2024), or exploratory and confirmatory factor analyses in instrument development and validation research (Lee and Park 2024). Experimental and quasi-experimental designs, though less prevalent, provide valuable insights into the effects of different pedagogies or interventions

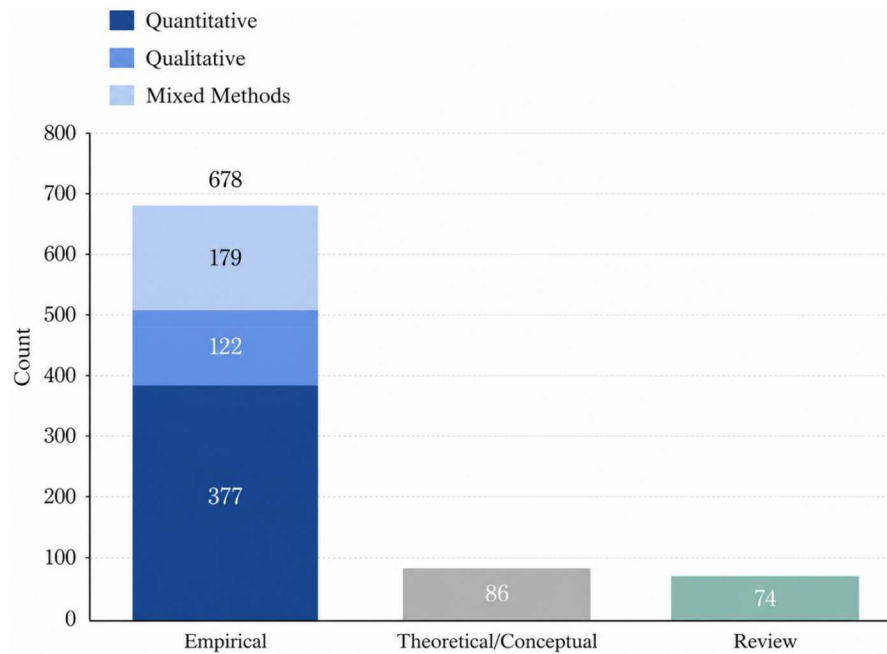


Figure 9. Methodological approaches and research design distribution in AI literacy research.

in developing students' AI literacy, including Dai's (2024) dual-contrast pedagogy and Joo and Park's (2024) AI-based convergence education programme targeting primary school students. Beyond these designs, person-centred and pattern-oriented analytical techniques have become more visible, including latent profile analysis (Wu, Wang, and Lalli 2026; Zhang and Kang 2026; Zhu et al. 2026), decision trees (Toker Gokce et al. 2025), and hierarchical cluster analysis (Lipovec 2024), as well as psychometric network analysis applied to two-wave AI literacy data (Shen et al. 2026), demonstrating growing methodological sophistication within the field.

5.2.2. Qualitative research and methodological approaches

Qualitative methodologies, while representing a smaller proportion of the research corpus, provide critical contextual depth through diverse approaches. Case studies constitute the most common qualitative strategy, offering rich descriptions of AI literacy development in naturalistic, classroom-based settings. These typically integrate multiple data-collection methods, including observations, interviews, and document analysis, to construct holistic understandings of AI literacy within specific contexts. Interview-based studies emphasise stakeholder perspectives, particularly among educators implementing AI literacy initiatives.

Emerging qualitative approaches include participatory design methods, exemplified by Strubberg, Bennett, and Nardone's (2024) study, which positioned students

in the pedagogy development for AI literacies, and Bilstrup et al.'s (2024) participatory infrastructure involving strategic partnership with nonprofit organisations and publicly funded newscasting companies to design and disseminate a web-based educational tool for building machine-learning models.

Notably, several qualitative studies employ innovative methodologies specifically adapted to AI literacy contexts, such as action research with youth (Aleman and Martinez 2024), design fiction (Xie et al. 2022) and narrative inquiry with adolescents (Yang and Hu 2025). These innovative qualitative techniques complement the commonly used ones, providing insights into AI literacy education in unique contexts, such as alternative school placement (Aleman and Martinez 2024), primary school students from marginalised backgrounds (Xie et al. 2022), and teenagers' lived experiences with conversational AI (Yang and Hu 2025).

Mixed-methods research (26.4% of empirical studies) demonstrates the field's growing recognition of the need to integrate quantitative measurement with qualitative contextualisation. These studies typically employ sequential designs, with quantitative measures (e.g. pre-post assessments, surveys) providing broad patterns that are subsequently explored through qualitative components (e.g. interviews, focus groups, reflective writing). This methodological integration is particularly evident in curriculum-evaluation research. For instance, Kong, Cheung, and Zhang's (2023) evaluation of AI literacy courses for university students from diverse backgrounds uses a combination of conceptual tests, surveys,

and self-reflective writing, and Bellas et al.'s (2024) long-term field study incorporates both survey data and detailed observational records. Zhang and Samsudin (2026) combine quasi-experiments with semi-structured interviews to evaluate AI-integrated pedagogy in EFL teacher education (Zhang and Samsudin 2026),

Several mixed-methods studies employ innovative data-integration approaches, such as workshop interventions with pre/post surveys and artifact analysis (Kaspersen et al. 2024) and quasi-experimental designs with qualitative user studies using interviews and recorded observations (Williams et al. 2024), and process-oriented designs that track the evolution of GenAI literacy across an academic term by integrating performance logs, survey data, and reflective writing (Yan, Nakajima, et al. 2025). The innovative methodological approaches and integration reflect growing methodological sophistication in the field.

The persistent emphasis on quantitative approaches, while enabling standardised measurement, statistical comparison, and broader generalizability across populations, continues to risk underrepresenting the contextual complexity and sociocultural dimensions of AI literacy development. In particular, survey-based assessments may overlook how learners interpret, negotiate, and apply AI-related knowledge within diverse educational, cultural, and technological contexts. Such approaches may also insufficiently capture the influence of learners' lived experiences, values, identities, and interactions with AI systems, all of which shape the development of AI literacy in meaningful ways.

However, the relatively substantial proportion of mixed-methods studies suggests that researchers are increasingly attempting to bridge this limitation by integrating quantitative measurement with qualitative inquiry. Rather than relying exclusively on either positivist or interpretive orientations, these studies seek to combine the strengths of statistical generalisation with richer contextual interpretation. This methodological shift reflects growing recognition that AI literacy is not merely a set of technical competencies that can be easily quantified, but also a socially situated and context-dependent construct that involves ethical awareness, critical reflection, and human-AI interaction.

Given the multidimensional nature of AI literacy, there is a need for complementary assessment approaches capable of capturing both measurable outcomes and experiential dimensions of learning. Combining questionnaires with interviews, reflective writing, observations, think-aloud protocols, or participatory methods may provide a more comprehensive understanding of how learners develop AI literacy

across different settings and populations. Such methodological diversity can also support the development of more theoretically grounded and contextually sensitive conceptualizations of AI literacy.

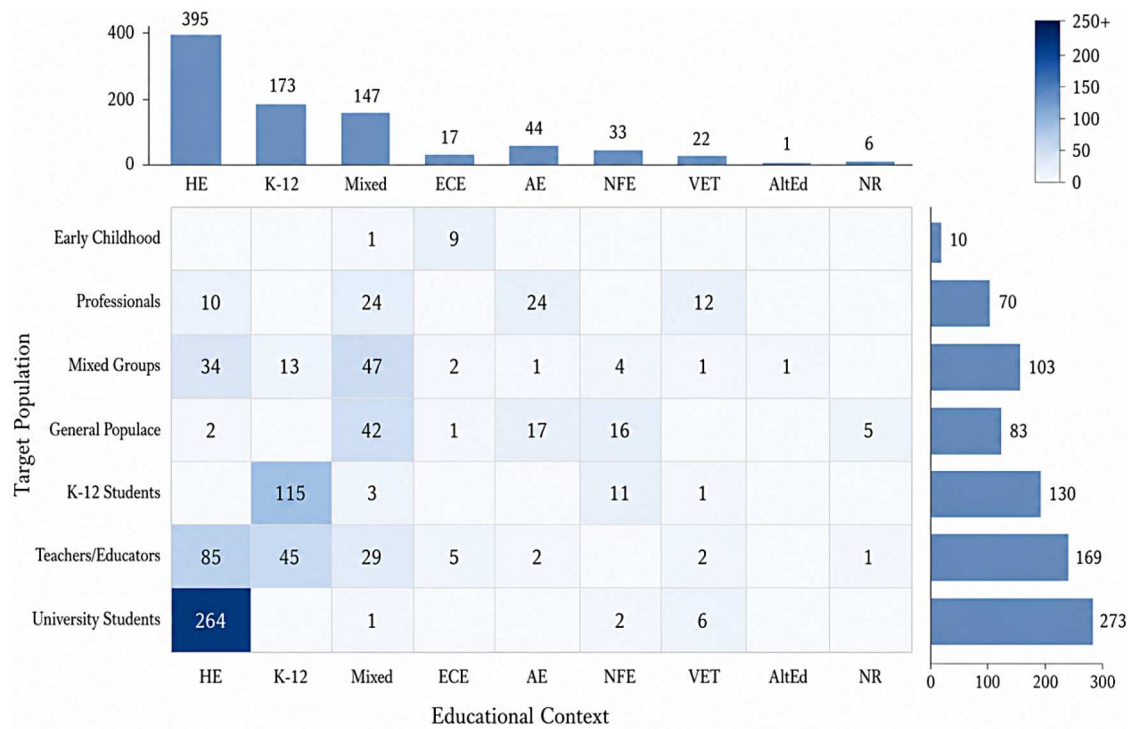
5.3. Target populations and educational contexts

Analysis of target populations and educational contexts in AI literacy research reveals distinctive patterns that influence the conceptualisation, implementation, and assessment of AI literacy initiatives. Examining these dimensions provides critical insights into how AI literacy is being conceptualised and operationalised across different educational settings and for diverse stakeholder groups, with implications for both theoretical development and educational practice.

As illustrated in Figure 10, the distribution of target populations is led by university students (32.6%), followed by teachers/educators (20.2%) and K-12 students (15.5%); these three groups together account for more than two-thirds of all studies. The remaining studies address more specialised populations, including mixed groups (12.3%), the general populace (9.9%), professionals in workplace contexts (8.4%), and young children in early childhood settings (1.2%). The clear pre-eminence of university students reflects both the accessibility of this population for empirical research and the perceived urgency of preparing graduates for AI-infused workplaces, while the substantial share of professionals indicates that workplace AI literacy has moved from a peripheral concern to a substantive research stream in its own right.

The distribution of educational contexts follows a similar pattern, with higher education institutions representing the strongly predominant research context (47.1%). The prevalence of higher education contexts may be attributed to advanced technological integration in higher education and perceived urgency of preparing graduates for AI-infused workplaces. Within higher education, AI literacy initiatives span diverse disciplines, such as business (Jang 2024), nursing (El-Sayed et al. 2025), library science (Hossain, Biswas, and Khan 2026), medical education (Laupichler et al. 2024), and chemistry (Young, Dawood, and Lewis 2024). Notably, discipline-specific adaptations emphasise contextual relevance, with Simms (2025) examining ethical implications in nursing education and Ou et al. (2024) conceptualising critical GenAI literacy for doctoral academic writing.

K-12 education constitutes the second-largest educational context (20.6%), with systematic reviews by Casal-Otero et al. (2023), Ng et al. (2021) and Yim and Su (2025) synthesising AI literacy education



Note. HE = higher education; K-12 = K-12 education; Mixed = mixed contexts; ECE = early childhood education; AE = adult education; NFE = non-formal education; VET = vocational education and training; AltEd = alternative education; NR = not reported.

Figure 10. Distribution of target populations and educational contexts in AI literacy research.

approaches across K-12 settings. This substantial representation demonstrates growing recognition of AI literacy's importance in compulsory education, where foundational competencies can be established early in students' educational trajectories. Notable K-12 research includes Dai (2024)'s experimental study of dual-contrast pedagogy for upper elementary students, Moreno-León et al. (2024)'s large-scale investigation of programming activities' impact on computational thinking and AI literacy in Spanish schools, and Ng et al.'s (2024c) examination of educational games to foster AI knowledge and engagement. The K-12 research corpus is characterised by distinct pedagogical emphases compared to higher education studies, with greater focus on age-appropriate conceptual development, gamified learning approaches, and integration with existing curricular frameworks. These studies typically employ scaffolded learning experiences that contextualise AI concepts within concrete applications relevant to children's daily experiences, as exemplified by Williams et al.'s (2024) development of the Doodlebot educational robot for fostering AI literacy through creative activities; and Lee et al.'s (2021) DAILY curriculum to develop secondary students' AI concepts, ethics, and AI related career awareness via online learning activities using digital tools.

Mixed educational contexts (17.5%) represent an important tertiary category, encompassing research spanning multiple educational levels or addressing cross-cutting literacy requirements. These studies frequently examine theoretical frameworks applicable across educational settings (Schüller 2022; Shiri 2024) or investigate pedagogical approaches with multi-level relevance (Bellás et al. 2024).

The remaining educational contexts, adult education (5.3%), non-formal education (3.9%), vocational education and training (2.6%), and early childhood education (2.0%), represent emerging or persistently under-served domains of AI literacy research, where specialised approaches are being developed for specific contexts.

Early childhood research, though limited, demonstrates promising innovations in age-appropriate AI literacy development. For instance, Su and Yang (2024) documented successful implementation of AI literacy curriculum for kindergarten students through embodied learning and play-based approaches; Lee, Ku, and Ko (2025) examined how AI robots can promote South Korean preschoolers' AI literacy and computational thinking; Collyer-Hoar et al. (2025) reported a co-design process for an inclusive AI literacy tool for children; and Lee (2026) proposed a

play-centred, design-based curriculum tailored to pre-K and kindergarten learners. Non-formal education research addresses settings such as museum exhibits (Long, Blunt, and Magerko 2021; Rollins et al. 2024) and family learning contexts (Druga, Christoph, and Ko 2022). Alternative education and unspecified educational contexts together account for less than 1% of studies, indicating significant gaps in the research landscape beyond conventional educational settings.

Cross-analysis of educational contexts and target populations reveals several salient patterns with implications for future research. First, early childhood remains the most severely underrepresented developmental stage in the corpus, with young children accounting for only 1.2% of target populations and early childhood education for 2.0% of contexts, proportions that have not improved alongside the field's overall expansion. The emergence of design-based and play-centred curricula for pre-K and kindergarten learners (Collyer-Hoar et al. 2025; Lee 2026; Lee, Ku, and Ko 2025) signals growing interest in this stage but has not yet constituted a developed research programme. Vocational education and training (2.6%) remains similarly under-served, despite the labour-market imperatives that AI integration places on technical and applied occupations.

Second, research in professional contexts has expanded substantially beyond the early cluster of theoretical and exploratory studies represented by Cetindamar et al. (2024), Kaspersen et al. (2024), Li and Kim (2024), and Mughari, Rafique, and Ali (2024), and now spans diverse occupational groups including nurses (Yu et al. 2026; Zeng et al. 2025), dermatologists (Wongvibulsin and Lee 2025), journalists (Wu 2025), auditors (Arianpoor, Borhani, and Safa 2026), knowledge workers (Zhang and Lee 2025), and academic librarians (Elsayed and Abusharhah 2025; Rafique, Mughari, and Khan 2025). The empirical breadth of this stream is, however, not matched by comparable theoretical development. AI literacy frameworks specifically tailored to adult professionals remain scarce, and most studies rely on instruments and constructs originally developed for student populations.

Third, the distribution analysis further reveals that AI literacy research predominantly addresses learning in formal educational settings, with limited attention to informal learning contexts or lifelong learning beyond institutional boundaries. As AI technologies increasingly permeate everyday environments, expanding research into non-formal learning contexts and lifelong AI literacy development represents a critical direction for advancing the field's conceptual frameworks and practical applications. Long, Blunt, and

Magerko (2021), Long et al. (2023) suggested that museum exhibits and home dialogues provide desirable informal learning opportunities for individuals to engage with interactive, informal activities to learn about AI concepts outside the traditional classroom setting. Hemment et al. (2023) showcased how artists used experiential and creative methods to enhance AI literacy for general populace. These studies point out the potential of community engagement and multidisciplinary collaborations in AI literacy research.

5.4. GenAI literacy: an emerging sub-domain

The rapid proliferation of GenAI technologies since late 2022 has catalysed GenAI literacy as a distinct research focus within the broader AI literacy landscape. As evidenced by the temporal keyword analysis (Section 4.5), GenAI emerged as the second-most-frequent keyword in 2023, with GenAI literacy, LLMs, and prompt engineering crystallizing as separate keywords by 2025, reflecting the field's responsiveness to the widespread adoption of LLMs following the public release of ChatGPT. This subsection examines how GenAI literacy has begun to take shape as a research sub-domain across conceptual, methodological, and applied dimensions.

Conceptual development has advanced along complementary directions. Cox (2024) reviewed definitions of algorithmic, AI, and GenAI literacy and proposed a responsible GenAI literacy construct extending beyond effective prompting to encompass critical evaluation of outputs, awareness of model limitations and biases, and ethical considerations specific to generative systems. Ou et al. (2024) advanced a critical GenAI literacy framework for doctoral academic writing, organised around interaction with GenAI tools, ethical aspects, technical values and limitations, and self-regulated learning. Mishra, Warr, and Islam (2023) extended TPACK to accommodate GenAI competencies in teacher knowledge structures. Together these frameworks suggest that GenAI literacy, while sharing foundations with broader AI literacy, requires additional dimensions, such as prompt formulation, hallucination detection, and responsible co-creation, which address the affordances and risks of generative systems.

Instrument development has expanded rapidly, with the corpus now containing 16 scales specifically measuring GenAI literacy. Beyond Lee and Park's (2024) early ChatGPT Literacy Scale, recent contributions include Wang, Wang et al.'s (2025; Wang, Derakhshan, et al. 2025) GenAI literacy scale validated with EFL teachers; Wu, Wang, and Lalli's (2026) GenAI Competence Scale combined with latent profile identification for pre-

service second-language teachers; Gümüş and Kara's (2025) Generative AI Literacy for Learning Scale; and Yan, Nakajima et al. (2025), which links GenAI literacy to privacy-protection and information-verification behaviours. A notable methodological advance is Jin et al.'s (2025) Generative AI Literacy Assessment Test, a 20-item multiple-choice instrument that measures performance rather than self-reported competence, partially addressing the self-report dependence identified in Section 5.1. Despite this proliferation, the theoretical relationship between GenAI literacy and broader AI literacy remains under-specified, and few scales have been comparatively evaluated across populations and languages.

Educational intervention studies have explored diverse pedagogical approaches for GenAI literacy development across disciplines and levels. Higher-education interventions span nursing education (Simms 2025), English for academic purposes (Ngo and Hastie 2025), marketing classrooms (Beninger et al. 2025), business communication (MacArthur et al. 2025), and pre-service language teacher education (Karaduman 2025). Secondary and primary applications include EFL academic writing (Woo et al. 2026) and STEAM-integrated GenAI literacy (Relmasira, Lai, and Donaldson 2023). A consistent characteristic across these studies is the emphasis on critical evaluation of AI-generated outputs and ethical reasoning, reflecting recognition that GenAI literacy extends beyond operational proficiency to informed judgement regarding output quality, accuracy, and appropriateness.

Complementing these intervention designs, relational studies have examined GenAI literacy's links with technology acceptance and behavioural intention (Al-Abdullatif and Alsubaie 2024; Jang 2024), adoption patterns and ethical perceptions (Chen, Tallant, and Selig 2025), and cross-national differences in student GenAI literacy (O'Dea et al. 2026). Process-oriented investigations such as Yan, Liu et al.'s (2025; Yan, Nakajima, et al. 2025) longitudinal tracking of GenAI literacy evolution across an academic term are beginning to complement variable-centred designs with developmental perspectives.

The substantial empirical research visible across conceptual, measurement, intervention, and relational strands indicates that GenAI literacy has moved from peripheral concern to substantive sub-domain. As GenAI technologies continue to evolve, the boundaries and core components of GenAI literacy will likely require ongoing reconceptualization. The theoretical relationship between GenAI literacy and broader AI literacy, whether the former constitutes a subset of the latter, an overlapping construct, or an emergent literacy

requiring distinct theoretical treatment, remains insufficiently clarified and warrants sustained scholarly attention.

6. Discussion

The integration of bibliometric analysis and thematic synthesis methods has enabled a more comprehensive understanding of the rapidly emerging field of AI literacy research than a single synthesis method. By examining both the structural patterns of knowledge production and the substantive content of scholarly contributions, this study provides a multidimensional perspective on the development and current state of this field. It also identifies research gaps and explores future opportunities from theoretical and methodological perspectives.

6.1. Theoretical gaps and opportunities

The analysis reveals a paradoxical state in AI literacy theory: rapid framework proliferation alongside limited theoretical consolidation. Long and Magerko's (2020) 17-competency model remains the most influential foundation, but the field has since witnessed the introduction of numerous alternative frameworks, such as Ng et al.'s (2021) four-aspect model and its ABCE refinement (Ng et al. 2024b), Wang, Rau, et al.'s (2023; Wang, Li, et al. 2023) four-dimensional measurement scale, Almatrafi, Johri, and Lee's (2024) six-construct synthesis, Cetindamar et al.'s (2024) employee-focused model, and Schüller's (2022) transdisciplinary data-AI framework.

As noted in Section 5.1, new frameworks are introduced more rapidly than existing ones are empirically tested, comparatively evaluated, or formally retired. The result is a theoretical landscape characterised more by accumulation than by integration, suggesting that synthetic theoretical work, identifying complementarities, overlaps, and disagreements among competing frameworks, now represents a higher-value contribution than yet another framework variant.

Despite diversification of frameworks, an emergent convergence is discernible. Across diverse frameworks, four dimensions consistently appear: technical understanding, practical application, critical evaluation, and ethical awareness. The temporal keyword analysis (Section 4.5) reinforces this convergence, with assessment instruments developed since 2023 typically operationalising these four broad areas regardless of terminological variation. This convergence offers a viable foundation for theoretical consolidation, though its translation

into a unifying framework would require systematic comparative analysis of how each dimension is defined and operationalised across populations and contexts, as well as deeper integration with established learning and literacy theories from broader educational research.

Another theoretical question raised by this synthesis concerns the relationship between AI literacy and GenAI literacy. As documented in Section 5.4, GenAI-related research has expanded rapidly, with dedicated conceptual frameworks (Cox 2024; Ou et al. 2024) and a growing set of GenAI literacy measurement instruments. Whether GenAI literacy represents a subset of AI literacy, an overlapping construct, or an emergent literacy requiring distinct theoretical treatment remains unresolved. If treated as a subset, existing AI literacy frameworks should be expanded to incorporate GenAI-specific competencies such as prompt formulation, hallucination detection, and responsible co-creation. If treated as a distinct construct, parallel theoretical development is warranted, with attention to its relationship to information literacy, media literacy, and emerging concerns about synthetic media. Resolving this question is urgent given the rapid pace of GenAI development and the accumulation of measurement instruments built on incompatible theoretical assumptions.

This synthesis reveals four further theoretical gaps that warrant attention. First, critical AI literacy remains underdeveloped relative to instrumental concerns. The temporal keyword analysis (Section 4.5) shows that acceptance, performance, and scale development dominate the field's vocabulary, while perspectives addressing algorithmic bias, social justice, and power asymmetries, articulated by Picasso et al. (2024) through a data-justice lens, complemented by Heyder and Posegga's (2021) explicit identification of functional, critical, and socio-cultural dimensions of AI literacy, remain peripheral rather than central to mainstream theorising. Systematic integration of these critical perspectives into core frameworks, rather than their treatment as a separate strand, is needed.

Second, the role of AI literacy in mitigating misinformation and synthetic-media risks represents an emerging but undertheorized priority. GenAI capabilities have dramatically lowered barriers to producing convincing synthetic text, images, and video (Lundberg and Mozellus 2025), yet existing AI literacy frameworks rarely articulate the specific competencies required to detect, evaluate, and respond to AI-generated misinformation. Theoretical work that bridges AI literacy with established media and information literacy traditions is needed to address this gap.

Third, inclusive AI literacy frameworks addressing diverse learning needs remain scarce. Current conceptualizations frequently assume neurotypical learners with reliable access to technological resources; leaving the competencies, supports, and adaptations needed by learners with disabilities, language minorities, and learners in low-resource contexts largely unspecified. Building inclusivity into frameworks at the conceptual stage, rather than as post-hoc accommodations, aligns with broader inclusive-literacy traditions (Fenwick and Comber 2021; Njenga, 2018) and is essential for equitable field development.

Fourth, the geographic and contextual concentration of theoretical production raises concerns about the cultural assumptions embedded in dominant AI literacy frameworks. As documented in Section 4.2, the United States, mainland China, Hong Kong, and Germany collectively contribute 51.4% of global output, while Africa, Latin America, and South Asia together account for a much smaller share of the corpus. Frameworks developed predominantly within high-resource Anglophone and East Asian contexts may not adequately reflect the socio-economic, linguistic, and infrastructural realities of less-resourced regions, with the attendant risk that one-size-fits-all models marginalise the very learners they aim to serve.

Comparable concerns apply to underrepresented developmental and professional contexts: early childhood, vocational education and training, and many adult professional settings remain inadequately theorised despite their importance to lifelong AI literacy development (Su and Yang 2022; Su, Ng, and Chu 2023). Co-developed, culturally responsive frameworks, produced in partnership with researchers, educators, and learners in underrepresented regions and developmental stages, represent both a theoretical and ethical imperative for the field's continued maturation.

6.2. Methodological gaps and opportunities

The methodological distribution among empirical studies: 55.6% quantitative, 26.4% mixed-methods, and 18.0% qualitative (Section 5.2), with the substantial mixed-methods share signalling growing recognition that AI literacy's multidimensional nature requires integrating statistical generalisation with contextual interpretation. Nevertheless, several limitations constrain the field's capacity to generate defensible knowledge.

The most consequential limitation is the near-universal reliance on cross-sectional designs. As documented in Section 5.1, most relational studies estimate associations between AI literacy and adjacent constructs

(self-efficacy, engagement, well-being, behavioural intention) using single-time-point data, leaving open whether AI literacy drives these outcomes, is shaped by them, or is jointly determined by unmeasured antecedents such as prior digital exposure or socio-economic background. Longitudinal designs that track AI literacy development over extended periods remain notably scarce, as do quasi-experimental designs capitalising on policy-driven exposure to AI literacy curricula, despite their importance for causal inference.

A second limitation concerns the dominance of self-report instruments. Most existing AI literacy scales only assess perceived rather than demonstrated competence. Although performance-based instruments have begun to emerge (notably Jin et al.'s (2025) GLAT and the small number of performance-oriented scales identified in Section 5.1), they remain rare relative to the volume of self-report-based empirical claims, weakening inferences about absolute proficiency levels and intervention effects.

Third, qualitative research (only 18.0% of empirical work), appears insufficient given the inherently situated nature of literacy development. Innovative qualitative designs, participatory design (Bilstrup et al. 2024; Strubberg, Bennett, and Nardone 2024), action research (Aleman and Martinez 2024), and narrative inquiry (Yang and Hu 2025), which well suit for experiential dimensions of human-AI interaction that variable-centred designs cannot capture, but they remain isolated cases rather than constitutive of a mature qualitative research programme.

Finally, design-based and process-oriented methodologies remain underdeveloped despite their suitability for theory-driven intervention development in authentic contexts. Promising examples include Rollins et al.'s (2024) iterative design of AI literacy museum exhibits and Yan, Liu, et al.'s (2025; Yan, Nakajima, et al. 2025) longitudinal process-oriented study of GenAI literacy development, alongside emerging applications of latent profile analysis (Pan and Wang 2025; Zhu et al. 2026) and psychometric network analysis (Shen et al. 2026) that capture heterogeneity and structural relationships within the construct. Sustained methodological pluralism, coupling robust psychometric work with longitudinal, qualitative, design-based, and person-centred approaches, will be essential for the field's continued maturation.

7. Conclusion

This study presents a comprehensive mapping of AI literacy research through an integrative review of 838 studies (2018–2026), revealing the intellectual landscape

and developmental trajectories of this rapidly emerging field.

7.1. Limitations of this synthesis

While this synthesis provides insights into AI literacy research, several limitations remain and offer directions for future work. First, the literature search was restricted to major academic databases and publications in English. Studies published in other languages or in less commonly indexed sources may have been overlooked, offering only a partial representation of the existing research. Future syntheses could broaden the scope by including multilingual sources and lesser-known databases to better capture contributions from underrepresented regions.

Second, the synthesis is constrained by the publication timeframe of the included studies. Given the rapid advancement of LLMs, GenAI, and agentic AI, AI literacy research is expanding at a fast pace; our synthesis only incorporates studies published up to early 2026, which limits coverage of the most recent developments. Ongoing reviews and updated syntheses will be needed to capture the field's continued evolution.

7.2. A future research agenda

Beyond the specific gaps detailed in Section 6, this synthesis suggests an ordered set of priorities for the field's next phase. The first-order priority is consolidating the field's theoretical and measurement infrastructure. Because the credibility of the rapidly expanding relational literature (30.5% of the corpus) rests on the constructs and instruments it employs, comparative studies that evaluate competing frameworks against empirical data, including testing the recurrent four-dimension structure of technical understanding, practical application, critical evaluation, and ethical awareness and establishing the discriminant validity of AI literacy relative to adjacent constructs, should now take precedence over further framework and scale proliferation. Two questions are particularly pressing within this agenda: (1) the theoretical relationship between AI literacy and GenAI literacy, which could be resolved by testing both types of instruments on the same samples; and (2) the systematic integration of critical perspectives on bias, equity, and power into core AI literacy frameworks, rather than their continued treatment as a peripheral strand.

A second priority is strengthening causal and developmental evidence. The field now possesses sufficient validated instruments and documented interventions

to support three key initiatives: tracking AI literacy longitudinally across educational transitions, replicating promising interventions across contexts with adequate comparison groups, and implementing performance-based outcome measurements. Such designs are needed to convert the current abundance of cross-sectional associations into defensible claims about how AI literacy develops and its real-world outcomes.

A third priority is deliberate diversification of research participation and coverage. The underrepresentation of early childhood, vocational, professional, and Global South contexts documented in Sections 4.2 and 5.3 is unlikely to correct itself through convenience-driven growth, which continues to concentrate on accessible university populations in high-resource regions. Progress will therefore require ecosystem-level effort, including cross-regional collaboration, capacity-building partnerships, and instruments and frameworks co-developing with researchers and educators in under-represented settings. Pursuing these priorities together will shift AI literacy research from a reactive model to a proactive one. Instead of constantly reacting to new technologies after they launch, researchers can build a forward-looking plan that anticipates future upgrades in generative and agentic AI.

Author contributions

CRediT: **Hao Zheng**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing; **Feifei Han**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Supervision, Validation, Writing – original draft, Writing – review & editing; **Hailiang Yang**: Formal analysis, Methodology, Resources, Software, Visualization, Writing – review & editing; **Jun Xu**: Investigation, Methodology, Software, Validation, Visualization, Writing – review & editing; **Yonghe Wu**: Funding acquisition, Investigation, Methodology, Writing – review & editing; **Jinran Wu**: Methodology, Validation, Visualization, Writing – review & editing; **Tong Li**: Methodology, Resources, Software, Validation, Visualization, Writing – review & editing.

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Appendix A. Complete search strings

Category	Search terms
AI literacy	'AI literacy' OR 'artificial intelligence literacy' OR 'AI competency' OR 'artificial intelligence competency' OR 'AI competencies' OR 'artificial intelligence competencies' OR 'AI capability' OR 'artificial intelligence capability' OR 'AI capabilities' OR 'artificial intelligence capabilities' OR 'AI skill' OR 'artificial intelligence skill' OR 'AI skills' OR 'artificial intelligence skills'
Generative AI literacy	'GenAI literacy' OR 'generative AI literacy' OR 'GenAI competency' OR 'generative AI competency' OR 'GenAI competencies' OR 'generative AI competencies' OR 'GenAI capability' OR 'generative AI capability' OR 'GenAI capabilities' OR 'generative AI capabilities' OR 'GenAI skill' OR 'generative AI skill' OR 'GenAI skills' OR 'generative AI skills'
GAI literacy	'GAI literacy' OR 'GAI competency' OR 'GAI competencies' OR 'GAI capability' OR 'GAI capabilities' OR 'GAI skill' OR 'GAI skills'
LLM literacy	'LLM literacy' OR 'large language model literacy' OR 'LLM competency' OR 'large language model competency' OR 'LLM competencies' OR 'large language model competencies' OR 'LLM capability' OR 'large language model capability' OR 'LLM capabilities' OR 'large language model capabilities' OR 'LLM skill' OR 'large language model skill' OR 'LLM skills' OR 'large language model skills'
ChatGPT literacy	'ChatGPT literacy' OR 'ChatGPT competency' OR 'ChatGPT competencies' OR 'ChatGPT capability' OR 'ChatGPT capabilities' OR 'ChatGPT skill' OR 'ChatGPT skills'
GPT literacy	'GPT literacy' OR 'GPT competency' OR 'GPT competencies' OR 'GPT capability' OR 'GPT capabilities' OR 'GPT skill' OR 'GPT skills'

Note: Search categories were combined using the Boolean operator OR. Web of Science: Topic Search (TS) field; Scopus: TITLE-ABS-KEY field.